BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 4 Packet.

Respiratory · Lymphatic · Digestive · Abdominal and Lower Limb Vessels · Endocrine

THIS PACKET	**Course Notes** 6 chapters, one section per competency, with the competency number on every section. Contents at the front, index at the back.
SEPARATE	**Lab Packet** The lab structure list, muscle and flow charts, and the questions you could be asked on the lab exam.
SEPARATE	**Note Sheets** One drawing sheet per competency, two prompts each. To check your drawing, scan the QR code on the sheet or open the note sheet key.

BIO 004 · MODULE 4

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Every chapter and every competency, with its page. The lab packet and the note sheets are separate documents.

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Everything else for Module 4

Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 4 COVERS

Module 4 is thirty six competencies, numbers 97 to 132 in your competency packet. Every section of these notes is headed with its competency number, so you can go from the packet to the page and back.

Respiratory opens the module, because it finishes the story the cardiovascular module started. The lymphatic system drains what the capillaries leave in the tissues. The digestive system takes food apart, and its accessory organs supply the tools. The vessels and nerves of the abdomen and lower limb are lab competencies, and endocrine anatomy closes the module.

This is anatomy. Where function appears in these notes it is there because the structure does not make sense without it.

CHAPTER 1

The Respiratory System

RESPIRATORY

COMPETENCIES 97 TO 104

A reference for the respiratory system video and lab. This chapter covers the upper and lower respiratory tracts, the larynx, the bronchial tree, the lungs and pleurae, the alveoli, and common respiratory disorders. The focus is on structure.

COMPETENCIES

1. **97** Upper and lower tracts, conducting and respiratory zones
2. **98** Nose, nasal cavity, and pharynx
3. **99** Larynx
4. **100** Trachea and bronchial tree
5. **101** Lungs and pleurae
6. **102** Respiratory zone and alveolar structure
7. **103** Thoracic wall and diaphragm
8. **104** Respiratory disorders by structure affected

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The respiratory system

drsrennie-stack.github.io/new-build-bio4-solano/respiratory-concept-videos.html

COMPETENCY 97 LECTURE AND LAB

Upper and lower tracts, conducting and respiratory zones

You should be able to: Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

The **respiratory system** is the set of organs that move air into and out of the body and bring that air close to the blood. Air follows one continuous path, from the nose to the tiny air sacs of the lungs. Anatomists group the parts of this path in two ways. One grouping is by location: the upper and lower respiratory tracts. The other grouping is by job: the conducting and respiratory zones. The two groupings use different dividing lines, so you need to learn both.

Upper and lower respiratory tracts

The **upper respiratory tract** is the nose, the nasal cavity, the paranasal sinuses, and the pharynx. These parts sit in the head and neck. The **lower respiratory tract** is the larynx, the trachea, the bronchi and all their branches, and the lungs. These parts sit in the neck and thorax. The larynx is the first structure of the lower tract, so the line between the two tracts falls where the pharynx meets the larynx. Some clinical sources count the larynx as part of the upper tract instead. These notes follow the anatomy textbook split.

Conducting and respiratory zones

The **conducting zone** is every passage that only carries air. It runs from the nose to the **terminal bronchioles**. Its walls are too thick for gases to cross into the blood. As air moves through it, the conducting zone warms, moistens, and filters the air. The **respiratory zone** is where gas exchange with the blood happens. It begins at the **respiratory bronchioles**, which have a few alveoli in their walls, and continues through the **alveolar ducts**, the **alveolar sacs**, and the **alveoli**. The line between the two zones lies deep inside the lung, where a terminal bronchiole gives way to a respiratory bronchiole.

Because the dividing lines are in different places, one structure can belong to the lower tract and the conducting zone at the same time. The trachea is a good example. Every part of the upper tract is in the conducting zone. The lower tract

holds the rest of the conducting zone plus the whole respiratory zone. Gas exchange itself is a physiology topic, so this chapter covers only the structures.

EACH MAJOR AIRWAY STRUCTURE BY TRACT AND ZONE

Structure	Tract	Zone
Nose and nasal cavity	Upper	Conducting
Paranasal sinuses	Upper	Air spaces that open off the nasal cavity. No gas exchange
Pharynx	Upper	Conducting
Larynx	Lower	Conducting
Trachea	Lower	Conducting
Main, lobar, and segmental bronchi	Lower	Conducting
Bronchioles and terminal bronchioles	Lower	Conducting
Respiratory bronchioles	Lower	Respiratory
Alveolar ducts and alveolar sacs	Lower	Respiratory
Alveoli	Lower	Respiratory

In lab you may be asked to trace one breath from the nostril to an alveolus. This is the full path in order.

1. **Nares** the nostrils, where air enters
2. **Nasal vestibule** the skin-lined entry, where hairs catch large particles
3. **Nasal cavity** air passes through the meatuses beside the conchae and is warmed and moistened
4. **Choanae** the back openings of the nasal cavity
5. **Nasopharynx** air only
6. **Oropharynx** shared with food
7. **Laryngopharynx** shared with food, where air turns forward into the larynx
8. **Larynx** air passes through the glottis, between the vocal folds
9. **Trachea** the windpipe, down to the carina
10. **Main bronchus** one into each lung
11. **Lobar bronchus** one to each lobe
12. **Segmental bronchus** one to each bronchopulmonary segment
13. **Bronchioles** small airways with no cartilage
14. **Terminal bronchiole** the last part of the conducting zone
15. **Respiratory bronchiole** the first part of the respiratory zone
16. **Alveolar duct** a passage lined with alveoli
17. **Alveolar sac** a cluster of alveoli around a shared space
18. **Alveolus** the air sac where gases cross into the blood

MEMORY TIP

Tract tells you where a structure is (upper or lower). Zone tells you what it does (carries air or exchanges gas).

COMPETENCY 98 LECTURE AND LAB

Nose, nasal cavity, and pharynx

You should be able to: Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

The nose is the main entrance for air. Besides carrying air, it warms, moistens, and filters the air, it houses the receptors for smell, and it acts as a resonating chamber for the voice. The **external nose** is the part you can see. Bone forms its upper part: the nasal bones, the frontal bone, and the maxillae. Flexible hyaline cartilage (the septal, alar, and lateral cartilages) forms its lower part, which is why the tip of the nose bends.

THE EXTERNAL NOSE

Structure	What it is
Root	The area between the eyebrows, where the nose meets the forehead
Bridge	The upper, bony part of the nose
Dorsum nasi	The anterior margin, running from the root to the apex
Apex	The tip of the nose
Alae	The flared sides of the nostrils
Nares	The nostrils, the external openings

The nasal cavity

The **nasal cavity** is the space behind the external nose. The **nasal septum** divides it into right and left halves. The front of the septum is **septal cartilage**. The back of the septum is bone: the **perpendicular plate of the ethmoid** above and the **vomer** below. The roof of the cavity is formed by the ethmoid and sphenoid bones. The floor is the palate. The **hard palate** is the bony front part of the floor. The **soft palate** is the muscular back part. The palate separates the nasal cavity from the mouth.

Just inside each nostril is the **nasal vestibule**. It is lined with skin, sebaceous glands, and coarse hairs that trap large particles. Three scroll-shaped shelves project from each lateral wall: the **superior nasal concha**, the **middle nasal concha**, and the **inferior nasal concha**. The superior and middle conchae are parts of the ethmoid bone. The inferior concha is a separate bone. The groove under each concha is a **meatus** (superior, middle, and inferior meatus). The conchae make the air tumble as it passes, so more of it touches the warm, moist lining. At the back of the cavity, the two **choanae** open into the nasopharynx.

PARTS OF THE NASAL CAVITY

Structure	What it is
Nasal septum	The midline wall: septal cartilage in front, the perpendicular plate of the ethmoid and the vomer behind
Nasal vestibule	The area just inside the nostrils, lined with skin, sebaceous glands, and coarse hairs that filter particles
Nasal conchae	The superior, middle, and inferior conchae on the lateral wall. They swirl the air and increase contact with the lining
Nasal meatuses	The air passages beneath each concha: superior, middle, and inferior
Hard palate	The bony front of the floor, separating the nasal cavity from the mouth
Soft palate	The muscular back of the floor. It lifts to close off the nasopharynx during swallowing

Structure	What it is
Choanae	The posterior openings where the nasal cavity becomes the nasopharynx

Two kinds of mucous membrane line the nasal cavity. The **olfactory mucosa** is a small patch in the roof of the cavity that holds the smell receptors. The **respiratory mucosa** lines the rest. Its epithelium is **pseudostratified ciliated columnar epithelium** with **goblet cells**. The goblet cells and glands make mucus that traps dust and microbes. The cilia sweep that mucus back toward the pharynx, where it is swallowed. A dense bed of capillaries close to the surface warms the incoming air, and it is also why nosebleeds (**epistaxis**) are common.

THE TWO MUCOSAE OF THE NASAL CAVITY

Mucosa	Epithelium	Role
Olfactory mucosa	Olfactory epithelium, in a small patch in the roof	Houses the receptors for smell
Respiratory mucosa	Pseudostratified ciliated columnar with goblet cells	Secretes mucus, traps particles, and sweeps them toward the pharynx to be swallowed

MEMORY TIP

Cold air makes the cilia sluggish, so mucus stops moving back toward the throat and drips out the front instead. That is why your nose runs in cold weather.

The paranasal sinuses

The **paranasal sinuses** are air-filled spaces inside four skull bones that surround the nasal cavity. Each sinus is named for the bone it sits in. They are lined with the same mucosa as the nasal cavity, and each one drains through a small opening into the nasal cavity. The sinuses make the skull lighter, help warm and moisten air, and add resonance to the voice.

THE FOUR PARANASAL SINUSES

Sinus	Bone	Where it sits	Drains into
Frontal sinus	Frontal bone	In the forehead, above the orbits	Middle meatus
Ethmoidal air cells	Ethmoid bone	Many small spaces between the nasal cavity and the orbits	Middle meatus (front and middle cells) and superior meatus (back cells)
Sphenoidal sinus	Sphenoid bone	Deep in the skull, behind the nasal cavity	The space above the superior concha (the sphenoethmoidal recess)
Maxillary sinus	Maxilla	In the cheek, below each orbit. The largest sinus	Middle meatus

The pharynx

The **pharynx** is the throat. It is a funnel-shaped muscular tube that runs from the base of the skull down to the level of the cricoid cartilage, where it continues as the esophagus. It connects the nasal cavity and mouth to the larynx and esophagus. It has three regions, top to bottom.

The **nasopharynx** sits behind the nasal cavity and above the soft palate. It carries air only. When you swallow, the soft palate and the **uvula** (the small fleshy tab hanging from the soft palate) lift to close it off, so food does not enter the nose. The **pharyngeal tonsil** (adenoid) sits on its back wall, and the **openings of the auditory tubes** sit on its side walls. The auditory tubes connect the nasopharynx to the middle ears. The **oropharynx** sits behind the mouth, from the soft

palate down to the epiglottis. It carries both air and food. The **palatine tonsils** sit in its side walls, and the **lingual tonsil** covers the back of the tongue. The **laryngopharynx** sits behind the larynx, from the epiglottis down to the esophagus. It also carries air and food. At its lower end the paths split: air goes forward into the larynx and food goes back into the esophagus.

THE THREE REGIONS OF THE PHARYNX, TOP TO BOTTOM

Region	Location	Carries	Epithelium and structures
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar. Holds the pharyngeal tonsil and the openings of the auditory tubes
Oropharynx	Behind the oral cavity, from the soft palate to the epiglottis	Air and food	Stratified squamous, which resists friction from food. Holds the palatine and lingual tonsils
Laryngopharynx	Behind the larynx, from the epiglottis to the esophagus	Air and food	Stratified squamous. The point where the air and food paths divide

The epithelium changes with the job. Where only air passes, the lining is ciliated and moves mucus. Where food also passes, the lining is stratified squamous, which stands up to rubbing.

COMPETENCY 99 LECTURE AND LAB

Larynx

You should be able to: Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

The **larynx**, or voice box, connects the laryngopharynx to the trachea. It sits in the front of the neck, at about the level of the third to sixth cervical vertebrae. It hangs from the **hyoid bone**, a U-shaped bone above it that is not part of the larynx itself. The larynx has three jobs. It holds the airway open, it keeps food and drink out of the lower airway, and it produces the voice.

The nine cartilages

Nine cartilages, joined by ligaments and muscles, build the larynx. Three are single (unpaired) and three come in pairs. The **thyroid cartilage** is the largest. It is shaped like a shield, and its front ridge is the **laryngeal prominence** (the Adam's apple). The **cricoid cartilage** sits just below it. It is shaped like a signet ring, wide at the back and narrow in front, and it is the only complete ring of cartilage in the airway. The trachea hangs from its lower edge. The **cricothyroid ligament** fills the gap between the thyroid and cricoid cartilages in front.

The **epiglottis** is a spoon-shaped flap attached to the inside of the thyroid cartilage. Its free top edge points up behind the tongue. During a swallow, the larynx rises and the epiglottis tips down over the **laryngeal inlet** (the opening into the larynx), so food slides past it into the esophagus. The paired **arytenoid cartilages** sit on top of the back of the cricoid. They anchor the vocal folds and pivot to move them, which makes them the most important paired cartilages. The small **corniculate cartilages** sit on top of the arytenoids. The small **cuneiform cartilages** lie in the fold of tissue that runs from each arytenoid up to the epiglottis.

THE NINE CARTILAGES OF THE LARYNX

Cartilage	Number	Cartilage type	Description
Thyroid cartilage	Single	Hyaline	The large shield-shaped cartilage. Its front ridge is the laryngeal prominence (Adam's apple)

Cartilage	Number	Cartilage type	Description
Cricoid cartilage	Single	Hyaline	The ring just below the thyroid cartilage. The only complete ring of cartilage in the airway
Epiglottis	Single	Elastic	The flexible spoon-shaped flap that tips over the laryngeal inlet during swallowing
Arytenoid cartilages	Paired	Mostly hyaline	Sit on the back of the cricoid. Anchor the vocal folds and move them
Corniculate cartilages	Paired	Elastic	Small cartilages on top of the arytenoids
Cuneiform cartilages	Paired	Elastic	Small cartilages in the fold between each arytenoid and the epiglottis

MEMORY TIP

The thyroid, cricoid, and arytenoid cartilages are hyaline cartilage (only the small tips of the arytenoids are elastic). The epiglottis is the one unpaired cartilage made of elastic cartilage, because it has to bend and spring back with every swallow. The tiny corniculate and cuneiform cartilages are elastic too.

The folds and the glottis

Two pairs of folds of mucous membrane stretch from front to back inside the larynx. The lower pair are the **vocal folds**, the true vocal cords. Each one contains a **vocal ligament**, an elastic band that runs from an arytenoid cartilage in back to the thyroid cartilage in front. Their lining has very few blood vessels, so they look pearly white. They vibrate as air passes between them, and that vibration makes sound. The upper pair are the **vestibular folds**, the false vocal cords. They help close the airway during swallowing and play no part in making sound.

The **glottis** is the vocal folds together with the opening between them. Books draw its edges a little differently: some use the name for the opening alone, and OpenStax also includes the vestibular folds. That opening (the rima glottidis) is the narrowest point that air passes through in the larynx. When the arytenoids pivot, the vocal folds swing apart to widen the opening for breathing or come together to close it.

THE FOLDS AND THE OPENING

Structure	What it is
Vocal folds	The true vocal cords, the lower pair. Pearly white, they vibrate as air passes to produce sound
Vocal ligaments	The elastic bands inside the vocal folds, running from the arytenoid cartilages to the thyroid cartilage
Vestibular folds	The false vocal cords, the upper pair. They help close the airway during swallowing and play no part in sound
Glottis	The vocal folds together with the opening between them

The lining changes at the vocal folds. Above them, where food can rub, the epithelium is stratified squamous. Below them, the lining is pseudostratified ciliated columnar epithelium, and its cilia sweep mucus up toward the pharynx.

COMPETENCY 100 LECTURE AND LAB

Trachea and bronchial tree

You should be able to: Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

Below the larynx the airway becomes a branching tree. The trunk is the trachea. Each branch divides into smaller branches, so the tubes get narrower and more numerous at every step. All of these tubes belong to the conducting zone, down to the terminal bronchioles.

The **trachea**, or windpipe, runs from the lower edge of the cricoid cartilage down into the mediastinum. It is about 10 to 12 cm long. Its wall holds 16 to 20 C-shaped rings of hyaline cartilage, the **tracheal cartilages**. The rings keep the trachea from collapsing. The open part of each C faces back, toward the esophagus. Smooth muscle, the **trachealis muscle**, bridges that gap. Because the back wall is soft, the esophagus can bulge forward a little when you swallow food. The trachea is lined with pseudostratified ciliated columnar epithelium with goblet cells. The cilia move mucus up toward the pharynx.

The lowest tracheal cartilage forms the **carina**, an internal ridge at the point where the trachea splits in two. This fork lies at about the level of the sternal angle (T4 to T5). The lining over the carina is very sensitive, and anything that touches it sets off a strong cough.

The trachea splits into the right and left **main bronchi** (primary bronchi). Each one enters a lung at the hilum. The **right main bronchus** is wider, shorter, and more vertical than the left. For that reason, an inhaled object usually lands in the right lung. Inside the lung, each main bronchus divides into **lobar bronchi** (secondary bronchi), one for each lobe. The right lung has three and the left has two. Each lobar bronchus divides into **segmental bronchi** (tertiary bronchi), one for each **bronchopulmonary segment**. Branching continues until the tubes are less than 1 mm across. These are **bronchioles**. The last and smallest of the conducting tubes are the **terminal bronchioles**.

1. **Trachea** the windpipe, held open by C-shaped rings of hyaline cartilage, with the trachealis muscle across the back
2. **Carina** the internal ridge where the trachea forks into the two main bronchi
3. **Main bronchi** the primary bronchi, one entering each lung. The right one is wider, shorter, and more vertical
4. **Lobar bronchi** the secondary bronchi, one to each lobe (three on the right, two on the left)
5. **Segmental bronchi** the tertiary bronchi, one to each bronchopulmonary segment
6. **Bronchioles** airways less than 1 mm across, with smooth muscle and no cartilage
7. **Terminal bronchioles** the last airways of the conducting zone. The next tubes, the respiratory bronchioles, start the respiratory zone

The wall changes as the tubes get smaller. The C-shaped rings of the trachea and main bronchi give way to irregular plates of cartilage in the smaller bronchi, and cartilage disappears completely in the bronchioles. The epithelium thins from pseudostratified columnar to simple columnar and then to simple cuboidal in the terminal bronchioles. Smooth muscle makes up a larger and larger share of the wall. Without cartilage to hold them open, the bronchioles narrow a great deal when their smooth muscle contracts.

HOW THE AIRWAY WALL CHANGES DOWN THE TREE

Level	Cartilage	Epithelium	Smooth muscle
Trachea	C-shaped hyaline rings	Pseudostratified ciliated columnar with goblet cells	Trachealis muscle across the back of each ring
Main bronchi	C-shaped rings	Pseudostratified ciliated columnar with goblet cells	A thin layer

Level	Cartilage	Epithelium	Smooth muscle
Lobar and segmental bronchi	Irregular plates	Pseudostratified ciliated columnar, with fewer goblet cells in the smaller branches	A more complete layer
Bronchioles	None	Simple columnar to simple cuboidal	A complete layer that makes up much of the wall
Terminal bronchioles	None	Simple cuboidal	A complete layer

MEMORY TIP

Going down the tree, cartilage shrinks and smooth muscle takes over. Rings, then plates, then none.

COMPETENCY 101 LECTURE AND LAB

Lungs and pleurae

You should be able to: Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

The two lungs fill most of the **thoracic cavity**. They sit on either side of the **mediastinum**, the central compartment that holds the heart, the great vessels, the trachea, and the esophagus. The diaphragm lies under them. Each lung is cone-shaped, soft, and spongy. The two lungs are not the same size or shape. The heart tilts to the left and takes up room on that side, so the left lung is smaller. The liver sits under the right side of the diaphragm and pushes it up, so the right lung is shorter and wider.

Surfaces and landmarks

The **apex** of each lung is its narrow top. It rises just above the clavicle, into the root of the neck. The **base** is the broad, curved bottom surface that rests on the diaphragm. The **costal surface** is the large outer surface that lies against the ribs. The **mediastinal surface** is the medial surface that faces the mediastinum. On it is the **hilum**, the indented area where the bronchus, blood vessels, nerves, and lymphatic vessels enter and leave the lung. Together, the structures passing through the hilum make up the **root of the lung**.

THE SURFACES AND LANDMARKS OF A LUNG

Structure	What it is
Apex	The narrow superior tip of a lung, rising just above the clavicle
Base	The broad inferior surface of a lung, resting on the diaphragm
Costal surface	The outer surface against the ribs
Mediastinal surface	The medial surface facing the heart and mediastinum
Hilum	The indentation on the mediastinal surface where the bronchus and vessels enter and leave
Root of the lung	The bundle at the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatic vessels
Cardiac notch	The indentation in the front edge of the left lung that makes room for the heart
Lingula	The tongue-shaped part of the left superior lobe, just below the cardiac notch

In the root of the lung, the **pulmonary artery** carries blood low in oxygen into the lung, and the **pulmonary veins** carry oxygen-rich blood back out to the heart. On the cadaver, the arrangement at the hilum helps you tell the right lung from the left. In the left hilum, the pulmonary artery is the highest structure. In the right hilum, a bronchus sits at or above the level of the artery. In both lungs, the pulmonary veins are the lowest structures.

Lobes and fissures

Deep grooves called **fissures** divide each lung into **lobes**. The right lung has three lobes: superior, middle, and inferior. Its **oblique fissure** separates the inferior lobe from the middle and superior lobes, and its **horizontal fissure** separates the superior lobe from the middle lobe. The left lung has two lobes, superior and inferior, separated by an oblique fissure only. The **lingula** of the left superior lobe sits in about the same place as the middle lobe on the right. Each lobe is further divided into bronchopulmonary segments, each served by its own segmental bronchus and blood vessels. The right lung has 10 segments, and the left has about 8 to 10.

COMPARE THE TWO LUNGS

Lung	Lobes	Fissures	Note
Right lung	Three: superior, middle, and inferior	A horizontal fissure and an oblique fissure	Larger. Shorter and wider, because the liver raises the diaphragm on the right
Left lung	Two: superior and inferior	An oblique fissure only	Smaller, with a cardiac notch for the heart and a lingula on the superior lobe

MEMORY TIP

Right has three, left has two. The left gave up its middle lobe to make room for the heart, and the lingula is what is left in its place.

The pleurae and the pleural cavity

Each lung is wrapped in its own double-layered serous membrane, the **pleura**. The **parietal pleura** lines the inside of the thoracic wall, the top of the diaphragm, and the sides of the mediastinum. The **visceral pleura** clings to the surface of the lung and dips into its fissures. The two layers are one continuous sheet. They turn back on each other at the hilum, where the root of the lung enters.

Between the two layers is the **pleural cavity**. It is a very thin space that holds only a film of **pleural fluid**. The fluid lets the lung glide against the chest wall as you breathe, and it holds the two layers together so the lung follows the chest wall as it expands. There is a right pleural cavity and a left pleural cavity, kept apart by the mediastinum, so air or fluid in one cavity does not collapse the other lung. At the bottom edge, where the parietal pleura turns from the ribs onto the diaphragm, it forms a low pocket called the **costodiaphragmatic recess**. This is the lowest part of the pleural cavity, so extra fluid collects there.

THE PLEURAE

Structure	What it is
Parietal pleura	The serous membrane lining the thoracic wall, the top of the diaphragm, and the mediastinum
Visceral pleura	The serous membrane covering the outer surface of the lung, including inside the fissures
Pleural cavity	The thin, fluid-filled space between the two pleurae. One on each side
Costodiaphragmatic recess	The lowest pocket of the pleural cavity, where the pleura turns from the ribs onto the diaphragm

COMPETENCY 102 LECTURE AND LAB

Respiratory zone and alveolar structure

You should be able to: Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

Past the terminal bronchioles, the airway becomes the respiratory zone. Here the walls thin out into tiny air sacs, and gases can move between the air and the blood. The respiratory zone is the bulk of the lung tissue. Under the microscope at low power it looks lacy, because it is mostly air spaces separated by thin walls.

1. **Respiratory bronchioles** the first airways of the respiratory zone, with a few alveoli budding from their walls
2. **Alveolar ducts** passages whose walls are lined entirely with alveoli
3. **Alveolar sacs** clusters of alveoli that open into a shared space, like a bunch of grapes
4. **Alveoli** the single tiny air sacs where gas exchange happens. The two lungs hold hundreds of millions of them

The alveolar wall

An **alveolus** is a small cup-shaped pocket of air with a wall only one cell thick. Most of that wall is made of **type I alveolar cells**. These are very thin, flat cells, a simple squamous epithelium, and they cover nearly all of the alveolar surface. Gases cross the wall through them. Scattered among them are **type II alveolar cells**. These are rounder, cuboidal cells that secrete **surfactant**, a fluid that lowers surface tension and keeps the alveoli from collapsing. Type II cells also secrete antimicrobial proteins.

Moving freely over the inner surface of the alveoli are the **alveolar macrophages**, also called dust cells. They engulf dust, microbes, and other debris that got past the conducting zone. Small openings called **alveolar pores** connect neighboring alveoli. Elastic fibers surround each alveolus and let the lung stretch on the way in and recoil on the way out.

The respiratory membrane and pulmonary capillaries

Each alveolus is wrapped in a dense net of **pulmonary capillaries**. These come from branches of the pulmonary arteries and drain into the pulmonary veins. Where a capillary presses against an alveolus, the air and the blood are separated by a very thin barrier called the **respiratory membrane**. It has three layers: the type I alveolar cell, the fused basement membranes of the alveolus and the capillary, and the capillary endothelial cell. Because the membrane is so thin and the alveoli are so many, the lungs have a huge surface for gas exchange.

THE PARTS OF THE ALVEOLUS

Structure	What it is
Type I alveolar cells	Thin, flat simple squamous cells that form most of the alveolar wall, where gases cross
Type II alveolar cells	Cuboidal cells scattered in the wall that secrete surfactant, which keeps the alveoli from collapsing
Alveolar macrophages	The dust cells that move over the alveolar surface and engulf debris and microbes
Alveolar pores	Small openings in the walls between neighboring alveoli
Respiratory membrane	The thin barrier between alveolar air and blood: type I cell, fused basement membranes, and capillary endothelium
Pulmonary capillaries	The dense net of capillaries wrapping each alveolus, fed by the pulmonary arteries and drained by the pulmonary veins

MEMORY TIP

Type I is thin, for exchange. Type II makes surfactant. Macrophages clean up.

COMPETENCY 103 LECTURE AND LAB

Thoracic wall and diaphragm

You should be able to: Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

The lungs sit inside a protective cage, the **thoracic wall**. It is built from the ribs and their costal cartilages, the sternum in front, and the thoracic vertebrae in back. Muscle fills the spaces between the ribs, and the diaphragm closes the bottom. Breathing works by changing the size of this cage, and the lungs follow the wall because the pleural fluid holds them to it.

The diaphragm

The **diaphragm** is a dome-shaped sheet of skeletal muscle that forms the floor of the thoracic cavity and separates it from the abdominal cavity. Its muscle fibers attach around the lower edge of the rib cage, the sternum, and the lumbar vertebrae, and they all pull toward a flat sheet of tendon at the top of the dome, the **central tendon**. The right dome sits a little higher than the left because the liver is under it. When the diaphragm contracts, it flattens and moves down, which makes the thoracic cavity taller. It is the main muscle of breathing.

The diaphragm is supplied by the right and left **phrenic nerves**. Each phrenic nerve comes from spinal nerves C3, C4, and C5 in the neck and travels down through the thorax to reach the diaphragm. On the cadaver, each phrenic nerve runs along the side of the pericardium, in front of the root of the lung.

MEMORY TIP

"C3, 4, and 5 keep the diaphragm alive." The phrenic nerve comes from those three spinal nerves.

The intercostal muscles

Two main layers of **intercostal muscles** fill each space between the ribs. They are supplied by the intercostal nerves. The **external intercostals** are the outer layer. Their fibers run downward and forward, the same direction as your fingers when your hands are in your front pockets. They lift the ribs during inspiration and work with the diaphragm in normal, quiet breathing. The **internal intercostals** lie just deep to them. Their fibers run at right angles to the external layer, downward and backward. They pull the ribs down during forced expiration, such as a hard cough or blowing out candles.

When you need to breathe in harder than normal, accessory muscles help lift the rib cage. These are the scalenes, the sternocleidomastoid, and the pectoralis minor. Quiet expiration needs no muscle effort, because the lungs and chest wall recoil on their own. Forced expiration uses the internal intercostals and the abdominal muscles.

MUSCLES OF THE THORACIC WALL AND DIAPHRAGM

Muscle	Fiber direction or shape	Role in breathing	Nerve
Diaphragm	Dome-shaped, fibers rising to the central tendon	Main muscle of inspiration. Contracts, flattens, and moves down	Phrenic nerves (C3 to C5)
External intercostals	Downward and forward, "hands in pockets"	Lift the ribs in inspiration, in quiet and forced breathing	Intercostal nerves
Internal intercostals	At right angles to the externals, downward and backward	Pull the ribs down in forced expiration	Intercostal nerves

Three structures through the diaphragm

Three large structures pass between the thorax and the abdomen through openings in the diaphragm. Each opening sits at a set vertebral level. The **inferior vena cava** passes through the **caval opening** in the central tendon at the level of the T8 vertebra. The **esophagus** passes through the **esophageal hiatus** in the muscle at T10, along with the vagus nerves. The **aorta** passes through the **aortic hiatus** at T12. This opening lies behind the diaphragm, between its two crura (the muscular legs that anchor it to the lumbar vertebrae). The thoracic duct passes through it with the aorta. Above the diaphragm the vessel is the thoracic aorta, and below it the vessel is called the abdominal aorta.

THE THREE OPENINGS, TOP TO BOTTOM

Vertebral level	Opening	Main structure	Also passes through
T8	Caval opening, in the central tendon	Inferior vena cava	
T10	Esophageal hiatus, in the muscle	Esophagus	Vagus nerves
T12	Aortic hiatus, behind the diaphragm between the crura	Aorta	Thoracic duct

MEMORY TIP

I-8, E-10, A-12. Inferior vena cava at T8, Esophagus at T10, Aorta at T12. The numbers are vertebral levels, and the structures are listed from top to bottom.

Other structures on the cadaver thorax

In lab, the thoracic dissection also shows the great vessels above the heart. For each one, say the side and whether it is an artery or a vein before you name it.

VESSELS YOU WILL FIND IN THE UPPER THORAX

Structure	Artery or vein	Where it is
Brachiocephalic trunk	Artery	First branch of the aortic arch, on the right only. It splits into the right subclavian and right common carotid arteries
Right and left common carotid arteries	Artery	Run up the neck to the head. The right comes from the brachiocephalic trunk, the left straight from the aortic arch
Right and left subclavian arteries	Artery	Run out to the upper limbs. The right comes from the brachiocephalic trunk, the left straight from the aortic arch
Right and left internal jugular veins	Vein	Drain the head and run down the neck beside the common carotid arteries
Right and left subclavian veins	Vein	Drain the upper limbs
Right and left brachiocephalic veins	Vein	Each forms where an internal jugular vein joins a subclavian vein. The two join to form the superior vena cava
Right and left pulmonary arteries	Artery	Carry blood from the heart into each lung at the hilum

Structure	Artery or vein	Where it is
Right and left pulmonary veins	Vein	Carry blood from each lung back to the heart

COMPETENCY 104 LECTURE

Respiratory disorders by structure affected

You should be able to: Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

Most respiratory disorders do one of three things. Some narrow or block the airways, so air has trouble moving in and out. Some damage or fill the alveoli, so the exchange surface is lost. Some affect the pleurae and pleural cavity, so the lung cannot stay against the chest wall. When you learn a disorder, start with the structure it affects and what happens to that structure.

COMMON RESPIRATORY DISORDERS BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What changes in that structure
Rhinitis	Nasal mucosa	The lining of the nasal cavity is inflamed, swollen, and makes extra mucus. Allergies or infections cause it
Sinusitis	Paranasal sinuses	The lining of the sinuses is inflamed, often when their openings into the nasal cavity are blocked
Laryngitis	Larynx and vocal folds	The vocal folds are inflamed and swollen, so they cannot vibrate normally and the voice turns hoarse
Asthma	Bronchi and bronchioles	The airway lining is inflamed, the smooth muscle tightens, and extra mucus collects, so the airways narrow
Chronic bronchitis	Bronchi	The bronchial lining stays inflamed and makes too much mucus, which blocks the airways
Cystic fibrosis	Bronchi and bronchioles	An inherited disorder in which the mucus is thick and sticky. The cilia cannot clear it, so it clogs the airways
Emphysema	Alveoli	The alveolar walls break down, and many small alveoli merge into fewer, larger spaces. The exchange surface shrinks and the lung loses its elastic recoil
COPD	Airways and alveoli	Chronic obstructive pulmonary disease, a long-term blockage of airflow. It is mainly emphysema and chronic bronchitis together
Pneumonia	Alveoli	An infection inflames the alveoli and fills them with fluid
Tuberculosis	Lung tissue	A bacterial infection that forms small nodules (tubercles) and scar tissue in the lung
Infant respiratory distress syndrome	Alveoli (type II alveolar cells)	A premature newborn's type II cells do not yet make enough surfactant, so the alveoli collapse with each breath
Lung cancer	Lung tissue	A malignant tumor that usually starts in the epithelium lining the airways or alveoli. Strongly linked to smoking

Disorder	Structure affected	What changes in that structure
Pleurisy	Pleurae	The pleural layers are inflamed and rough, so they rub against each other and cause sharp pain with each breath
Pleural effusion	Pleural cavity	Extra fluid collects in the pleural cavity and presses on the lung. In a person sitting or standing, it pools in the costodiaphragmatic recess
Pneumothorax	Pleural cavity	Air enters the pleural cavity, the layers separate, and the lung on that side collapses

Notice how the anatomy explains each problem. Asthma narrows the bronchioles so much because they have no cartilage to hold them open. Emphysema and pneumonia both take away working alveoli. A pneumothorax affects only one lung, because each lung has its own pleural cavity.

WHY THE DESIGN HAS WEAK POINTS

The **epiglottis** normally covers the laryngeal inlet during a swallow. When something slips past it, a coughing fit is the airway's attempt to clear it. Because the right main bronchus is wider, shorter, and more vertical than the left, an inhaled object most often lands in the right lung.

The **alveoli** give the lungs an enormous surface, but that surface is fragile. Emphysema destroys alveolar walls and merges many small sacs into a few large ones, trading a huge exchange surface for a small one. The change is structural and permanent.

The **pleural cavity** is sealed and holds only a thin film of fluid. If the chest wall or the lung is punctured, air enters that space, the layers separate, and the lung collapses. This is a pneumothorax. The same thin, sealed space that lets the lung glide is what makes it vulnerable.

The **cricothyroid ligament** spans the gap between the thyroid cartilage and the cricoid cartilage, the only complete ring in the airway. It lies just under the skin with no cartilage over it, below the vocal folds. That is why it is the landmark for an emergency airway when the larynx above it is blocked.

CHAPTER 2

The Lymphatic System

LYMPHATIC

COMPETENCIES 105 TO 108

A reference for the lymphatic system video and lab. This chapter covers lymph and the pathway it follows, the lymphatic vessels, lymph nodes, the lymphatic organs, and common disorders of the system.

COMPETENCIES

1. **105** Lymph pathway
2. **106** Vessel and node structure
3. **107** Primary and secondary organs
4. **108** Lymphatic disorders

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The lymphatic system

drsrennie-stack.github.io/new-build-bio4-solano/lymphatic-concept-videos.html

COMPETENCY 105 LECTURE

Lymph pathway

You should be able to: Define lymph and trace its one-way route from lymphatic capillaries through collecting vessels, nodes, trunks, and ducts back to the subclavian veins.

Blood capillaries leak. All day long, fluid filters out of them into the spaces between cells. This fluid is called **interstitial fluid**. The blood capillaries take most of it back, but some fluid and a small amount of plasma protein stay behind in the tissues. The lymphatic system collects this leftover fluid, about 3 liters a day, and returns it to the blood. This keeps fluid from building up in the tissues and keeps blood volume steady.

Once interstitial fluid enters a lymphatic vessel, it is called **lymph**. Lymph is clear and watery. It carries water, plasma proteins, and whatever else it picks up in the tissues, such as cell debris, bacteria, and sometimes cancer cells. Lymph coming from the small intestine after a fatty meal looks milky white. This fat-rich lymph is called **chyle**.

THE FLUIDS AND THE SYSTEM

Term	What it is
Lymphatic system	A network of vessels, tissues, and organs that drains tissue fluid and houses the body's defenses
Interstitial fluid	The fluid in the spaces between cells, filtered out of the blood capillaries
Lymph	Interstitial fluid once it has entered a lymphatic vessel; clear and watery
Chyle	Milky, fat-rich lymph coming from the lacteals of the small intestine
Three functions	Returns excess interstitial fluid to the blood, absorbs dietary fats from the small intestine, and houses the immune defenses

The lymphatic system is a one-way system. It does not form a loop the way blood vessels do. Lymph starts in the tissues and flows only toward the heart. As it travels, small vessels join into larger and larger vessels, the way small streams join to form a river. On the way, the lymph passes through lymph nodes, which filter it before it reaches the blood. Follow the route in order.

1. **Lymphatic capillaries** Blind-ended tubes in the tissues take up excess interstitial fluid. The fluid is now lymph.
2. **Lymphatic collecting vessels** Larger vessels with valves carry the lymph onward and keep it moving one way.
3. **Lymph nodes** Nodes spaced along the collecting vessels filter the lymph. Lymph usually passes through several nodes in a row.
4. **Lymphatic trunks** Collecting vessels merge into large trunks. Each trunk drains one major region of the body.
5. **Lymphatic ducts** The trunks empty into one of two ducts, the right lymphatic duct or the larger thoracic duct.
6. **Subclavian veins** Each duct empties into the blood where the subclavian vein meets the internal jugular vein on its own side.

Trunks and ducts

There are nine **lymphatic trunks**. Each one is named for the region it drains. Four of them come in left and right pairs, and one, the intestinal trunk, is single.

THE NINE TRUNKS, WHAT EACH DRAINS, AND WHERE IT EMPTIES

Trunk	Region drained	Empties into
Lumbar (left and right)	Lower limbs, pelvis, and abdominal wall	Thoracic duct, at the cisterna chyli
Intestinal (single)	Digestive organs of the abdomen	Thoracic duct, at the cisterna chyli
Bronchomediastinal (left and right)	Thoracic organs and the thoracic wall	Left: thoracic duct. Right: right lymphatic duct
Subclavian (left and right)	Upper limbs	Left: thoracic duct. Right: right lymphatic duct
Jugular (left and right)	Head and neck	Left: thoracic duct. Right: right lymphatic duct

The **right lymphatic duct** is short and small. It drains the right upper limb and the right side of the head, neck, and thorax. The **thoracic duct** is the largest lymphatic vessel in the body. It drains everything else, which is about three quarters of the body. The thoracic duct begins in the abdomen as a sac called the **cisterna chyli**, which sits in front of the upper lumbar vertebrae. From there it passes up through the diaphragm with the aorta and climbs through the thorax beside the vertebral column.

Both ducts end in the base of the neck. Each one empties into the junction of the **subclavian vein** and the **internal jugular vein** on its own side. The right lymphatic duct empties on the right side. The thoracic duct empties on the left side. At this point lymph becomes part of the blood plasma again.

THE RIGHT LYMPHATIC DUCT COMPARED WITH THE THORACIC DUCT

Feature	Right lymphatic duct	Thoracic duct
Size	Short and small	Largest lymphatic vessel
Drains	Right upper limb and the right side of the head, neck, and thorax	Everything else, about three quarters of the body
Trunks it receives	Right jugular, right subclavian, right bronchomediastinal	Both lumbar, intestinal, left jugular, left subclavian, left bronchomediastinal
Starts at	The right side of the base of the neck	The cisterna chyli, a sac in the abdomen

Feature	Right lymphatic duct	Thoracic duct
Empties into	Junction of the right subclavian and right internal jugular veins	Junction of the left subclavian and left internal jugular veins

How lymph moves without a pump

The lymphatic system has no heart to push the lymph. Lymph is moved by forces from the structures around the vessels, and the valves keep it from sliding backward.

- Skeletal muscle pump: contracting muscles squeeze the vessels that run between them.
- Respiratory pump: pressure changes in the thorax with each breath pull lymph upward.
- Smooth muscle in the walls of the larger lymphatic vessels contracts in rhythm.
- Pulsing of nearby arteries presses on the vessels.
- Valves in the collecting vessels keep all of this movement going one way, toward the heart.

MEMORY TIP

Split the body into four quarters. The right upper quarter (right arm, right side of the head, neck, and chest) drains into the right lymphatic duct. The other three quarters drain into the thoracic duct.

COMPETENCY 106 LECTURE AND LAB

Vessel and node structure

You should be able to: Describe the structure of lymphatic capillaries, valved collecting vessels, and a lymph node including capsule, cortex, medulla, hilum, and afferent and efferent vessels.

Lymphatic capillaries

Lymphatic capillaries are the smallest lymphatic vessels. They lie in the tissues, woven between the blood capillaries. Each one is a thin tube that is closed at one end, so it is called blind-ended. Its wall is a single layer of endothelial cells.

The endothelial cells of a lymphatic capillary overlap at their edges. These loose, **overlapping endothelial cells** act as one-way flap valves. When fluid builds up in the tissue, the flaps push open and fluid flows in. When pressure inside the capillary rises, the flaps press shut, so the fluid cannot flow back out. Fine collagen strands called **anchoring filaments** tie the endothelial cells to the surrounding tissue. When the tissue swells, the filaments pull on the cells and hold the capillary open instead of letting it collapse.

Because the gaps between the cells are large, lymphatic capillaries are more permeable than blood capillaries. Proteins, cell debris, bacteria, and even cancer cells can get in. Lymphatic capillaries are found in nearly every tissue. They are absent from the central nervous system (the brain and spinal cord), bone and red bone marrow, and avascular tissues such as cartilage, the epidermis, and the cornea.

LYMPHATIC CAPILLARIES COMPARED WITH BLOOD CAPILLARIES

Feature	Lymphatic capillary	Blood capillary
Ends	Blind (closed) at one end	Open at both ends, part of a loop
Size	Wider	Narrower
Wall	Overlapping endothelial cells act as one-way flap valves	Endothelial cells joined more tightly
Held open by	Anchoring filaments tied to the surrounding tissue	Pressure of the blood inside

Feature	Lymphatic capillary	Blood capillary
What gets in	Fluid, proteins, cells, pathogens, and even cancer cells	Mostly small molecules
Flow	One way, away from the tissues	Through, from the artery side to the vein side

In the small intestine, each villus has a special lymphatic capillary in its center called a **lacteal**. Lacteals absorb dietary fats, which are packaged in particles too large to enter blood capillaries. The fat turns the lymph milky, and this lymph (chyle) drains to the cisterna chyli and up the thoracic duct.

Collecting vessels and their valves

Lymphatic capillaries join to form **lymphatic collecting vessels**. These are built like small veins. Their walls have the same three layers as a vein, but the layers are thinner. Collecting vessels have more internal valves than veins do. These **lymphatic valves** are one-way flaps, so lymph can move forward but cannot flow backward. When a collecting vessel is full, the stretches between valves bulge, which gives the vessel a beaded look.

Superficial collecting vessels run in the subcutaneous layer of the skin alongside the superficial veins. Deep collecting vessels run alongside the deep arteries. Along their length, the collecting vessels pass through lymph nodes.

The lymph node

A **lymph node** is a small, bean-shaped organ that filters lymph. Hundreds of nodes lie along the collecting vessels. They are clustered in the neck (**cervical nodes**), the armpit (**axillary nodes**), and the groin (**inguinal nodes**), and there are many more deep in the thorax and abdomen.

The outside of a node is a **capsule** of dense connective tissue. Strands of connective tissue called **trabeculae** extend inward from the capsule and divide the node into compartments. Inside, a mesh of reticular fibers holds the lymphocytes and macrophages in place.

The node has two regions. The outer region is the **cortex**. The outer cortex holds rounded clusters of lymphocytes called **lymphatic follicles**, which are mostly B cells. Many follicles have a pale **germinal center** where B cells multiply. The deep cortex is mostly T cells. The inner region is the **medulla**. It contains **medullary cords**, strands of lymphocytes and plasma cells, separated by **medullary sinuses**, open channels lined with macrophages.

One side of the node is indented. This is the **hilum**. Blood vessels enter and leave the node at the hilum, and lymph leaves there too. **Afferent lymphatic vessels** carry lymph into the node. There are several of them, and they enter at many points on the rounded (convex) surface through the capsule. **Efferent lymphatic vessels** carry filtered lymph out of the node, and they leave only at the hilum. Because there are more afferent vessels than efferent vessels, lymph backs up and moves slowly through the node. This gives the macrophages and lymphocytes time to work.

THE PARTS OF A LYMPH NODE

Structure	What it is
Capsule	Dense connective tissue covering; trabeculae extend inward from it
Afferent lymphatic vessels	Several vessels that carry lymph into the node through the capsule, on the convex surface
Subcapsular sinus	The space just under the capsule where lymph first enters
Cortex	Outer region; the outer cortex has lymphatic follicles of B cells with germinal centers, and the deep cortex is mostly T cells
Medulla	Inner region of medullary cords and medullary sinuses lined with macrophages
Hilum	The indented side of the node, where blood vessels enter and leave and efferent vessels leave

Structure	What it is
Efferent lymphatic vessels	Fewer vessels that carry filtered lymph out of the node at the hilum

1. **Afferent lymphatic vessels** Lymph enters through the capsule on the convex surface.
2. **Subcapsular sinus** Lymph spreads out in the space just under the capsule.
3. **Cortical sinuses** Lymph seeps through the cortex, past the follicles.
4. **Medullary sinuses** Macrophages lining these channels engulf bacteria, debris, and cancer cells.
5. **Efferent lymphatic vessels** Filtered lymph leaves at the hilum and moves on to the next node or a trunk.

MEMORY TIP

Afferent vessels arrive, efferent vessels exit. The same two words mean "in" and "out" in other systems too, such as the arterioles of the kidney and the neurons of the nervous system.

COMPETENCY 107 LECTURE

Primary and secondary organs

You should be able to: Sort red bone marrow, thymus, lymph nodes, spleen, tonsils, and MALT into primary and secondary lymphoid organs with location and role.

The main cells of the lymphatic system are **lymphocytes**, a type of white blood cell. The two kinds you need here are **B cells** and **T cells**. All lymphocytes are made in the red bone marrow. Before they can fight anything, they have to finish maturing. B cells mature in the red bone marrow. T cell precursors leave the marrow and travel in the blood to the thymus, where they mature.

This gives a simple way to sort the lymphoid (lymphatic) organs. **Primary lymphoid organs** are where lymphocytes are made and mature. There are only two: the red bone marrow and the thymus. **Secondary lymphoid organs** are where mature lymphocytes gather and meet pathogens. These are the lymph nodes, the spleen, the tonsils, and MALT. Each secondary organ sits where it can catch pathogens from one source: the lymph nodes check lymph, the spleen checks blood, and the tonsils and MALT guard the linings of open tracts.

THE PRIMARY AND SECONDARY LYMPHOID ORGANS

Organ	Group	Location	Role
Red bone marrow	Primary	Spongy bone of the axial skeleton, the limb girdles, and the proximal ends of the femur and humerus	Where all lymphocytes form, and where B cells mature
Thymus	Primary	Mediastinum, above the heart and behind the sternum	Where T cells mature
Lymph nodes	Secondary	Along the collecting vessels, clustered in the cervical, axillary, and inguinal regions	Filter the lymph
Spleen	Secondary	Upper left abdomen, to the left of and behind the stomach, under the lower left ribs	Filters the blood
Tonsils	Secondary	A ring of lymphatic tissue in the walls of the pharynx and at the base of the tongue	Guard against inhaled and swallowed pathogens
MALT	Secondary	Mucous membranes of the digestive, respiratory, urinary, and reproductive tracts	Guards the mucosal surfaces

MEMORY TIP

Only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. B cells mature in the bone marrow and T cells mature in the thymus. Everything else on the list is secondary.

The thymus

The **thymus** sits in the mediastinum, above the heart and just behind the sternum. It is large in infants and children. After puberty it shrinks, and in older adults it is mostly fat. Its outer cortex is packed with immature T cells. Its inner medulla has fewer, more mature T cells and small round structures called thymic corpuscles. The thymus does not filter lymph. Unlike a lymph node, it has no afferent lymphatic vessels and no follicles.

The spleen

The **spleen** is the largest lymphatic organ. It sits in the upper left abdomen against the diaphragm, to the left of and behind the stomach, and it is protected by the lower left ribs. The splenic artery and splenic vein pass through its hilum. The spleen filters the blood. It removes worn-out red blood cells and platelets and responds to pathogens carried in the blood. Its tissue comes in two kinds.

THE SPLEEN AND ITS TWO TISSUES

Structure	What it is
Spleen	The largest lymphatic organ, in the upper left abdomen under the lower left ribs
White pulp	Small islands of lymphocytes clustered around central arteries; carries out the spleen's immune functions
Red pulp	Most of the organ; venous sinuses and splenic cords full of red blood cells and macrophages that filter the blood and remove worn-out erythrocytes

The tonsils and MALT

MALT stands for mucosa-associated lymphatic tissue. It is lymphatic tissue in the mucous membranes that line the open tracts of the body, where pathogens can get in. The largest and best organized pieces of MALT are the tonsils, the Peyer patches, and the appendix.

The **tonsils** form a ring of lymphatic tissue around the entrance to the pharynx, where they meet inhaled and swallowed material. Their surface has deep pits called crypts that trap bacteria and debris.

THE TONSILS AND OTHER MALT

Structure	Location	Notes
Pharyngeal tonsil	Posterior wall of the nasopharynx	Single; also called the adenoid
Palatine tonsils	Lateral walls of the oropharynx, between the palatoglossal and palatopharyngeal arches	Paired and the largest; the ones usually infected and removed
Lingual tonsils	Base (posterior third) of the tongue, facing the oropharynx	Guard swallowed material
Tubal tonsils	Around the openings of the auditory tubes in the nasopharynx	Paired
Peyer patches	Wall of the ileum, the last part of the small intestine	Clusters of lymphatic follicles that watch the intestinal contents
Appendix	Off the cecum of the large intestine	Wall packed with lymphatic follicles

COMPETENCY 108 LECTURE

Lymphatic disorders

You should be able to: Name edema, lymphedema, lymphadenopathy, lymphoma, splenomegaly, and tonsillitis and identify the structure each one affects.

Lymphatic disorders fall into two groups. In the first group, drainage fails and fluid collects in the tissues, which shows up as swelling. In the second group, the lymphatic tissue itself changes. A node, the spleen, or the tonsils become enlarged, inflamed, or cancerous.

Edema is swelling caused by excess fluid in the interstitial space. It happens when fluid leaks out of the blood capillaries faster than the lymphatic vessels can carry it away. **Lymphedema** is a specific kind of edema. It is long-term swelling caused by blocked, damaged, or missing lymphatic vessels. It can follow surgery that removes lymph nodes, because the vessels that passed through those nodes no longer drain the region.

Lymphadenopathy means enlarged lymph nodes. Nodes often swell when they are fighting an infection nearby, and they can also enlarge when cancer cells lodge in them. **Lymphoma** is a cancer of the lymphocytes in lymphatic tissue. It usually starts in lymph nodes and comes in two main groups, Hodgkin lymphoma and non-Hodgkin lymphoma. **Splenomegaly** is an enlarged spleen. It can happen with an infection such as mononucleosis or with a blood disorder such as leukemia. **Tonsillitis** is inflammation of the tonsils, usually the palatine tonsils, and it is common in childhood.

LYMPHATIC DISORDERS BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What changes
Edema	The interstitial space	Excess interstitial fluid builds up and the tissue swells, because fluid leaks out faster than the lymphatics can drain it
Lymphedema	The lymphatic vessels	Blocked, damaged, or missing vessels cause long-term swelling of the region they drain, sometimes after lymph node removal
Lymphadenopathy	The lymph nodes	The nodes enlarge, often while fighting an infection, and sometimes because cancer cells have lodged in them
Lymphoma	Lymphatic tissue, usually the lymph nodes	Lymphocytes become cancerous and the tissue enlarges; Hodgkin or non-Hodgkin lymphoma
Splenomegaly	The spleen	The spleen enlarges, for example in mononucleosis or leukemia, and becomes easier to rupture
Tonsillitis	The tonsils, usually the palatine tonsils	The tonsils become infected, red, and swollen

THE SYSTEM A CLINICIAN READS THROUGH THE SKIN

Swollen glands are **lymphadenopathy**. Lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.

Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the **sentinel node**, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.

The **spleen** sits just under the lower left ribs. Its soft tissue tears easily, and it holds a large volume of blood, so a blow to that area can rupture the spleen and cause heavy internal bleeding. An enlarged spleen, as in mononucleosis, ruptures even more easily.

CHAPTER 3

The Alimentary Canal

DIGESTIVE

COMPETENCIES 109 TO 119

Covers the wall of the digestive tube, the peritoneum, and the regions of the canal from the mouth to the anus, with the named features of the stomach and the intestines. The focus is on the structures and the job each one does.

COMPETENCIES

1. **109** Four-layer gut wall
2. **110** Enteric nerve plexuses
3. **111** Peritoneum and folds
4. **112** Retroperitoneal organs
5. **113** Mouth, pharynx, esophagus
6. **114** Major canal sphincters
7. **115** Stomach regions and features
8. **116** Small intestine and surface area
9. **117** Large intestine features
10. **118** Trace the pathway of food
11. **119** Clinical anatomy of canal disorders

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The digestive system

drsrennie-stack.github.io/new-build-bio4-solano/gi-concept-videos.html

COMPETENCY 109 LECTURE AND LAB

Four-layer gut wall

You should be able to: Name the four layers of the alimentary canal wall from lumen outward and state the dominant tissue of each.

The digestive system has two groups of organs. The **alimentary canal**, also called the gastrointestinal (GI) tract, is one continuous muscular tube that runs from the mouth to the anus. Food travels through it. The **accessory digestive organs** (the teeth, tongue, salivary glands, liver, gallbladder, and pancreas) sit beside the tube and empty their secretions into it. They have their own chapter.

The organs of the canal are the mouth, the pharynx, the esophagus, the stomach, the small intestine, and the large intestine. The open space inside the tube is the **lumen**. The lumen is open to the outside world at the mouth and at the anus.

From the esophagus to the anal canal, the wall of the tube has the same four layers. From the lumen outward they are the **mucosa**, the **submucosa**, the **muscularis** (also called the muscularis externa), and the **serosa**. Each region changes the details of these layers to suit its job, but the order never changes.

THE FOUR LAYERS OF THE WALL, FROM THE LUMEN OUTWARD

Layer	Dominant tissue	What it contains

Layer	Dominant tissue	What it contains
1. Mucosa	Epithelium, with a thin connective tissue layer and a thin smooth muscle layer beneath it	Three parts: the **epithelium** that touches the food; the **lamina propria**, a loose (areolar) connective tissue with capillaries and lymphatic tissue; and the **muscularis mucosae**, a thin layer of smooth muscle
2. Submucosa	Areolar connective tissue (OpenStax describes it as dense connective tissue)	Larger blood vessels and lymphatic vessels, and the submucosal nerve plexus
3. Muscularis	Smooth muscle in most of the tube	An inner circular layer and an outer longitudinal layer, with the myenteric nerve plexus between them. Skeletal muscle replaces smooth muscle in the mouth, pharynx, upper esophagus, and external anal sphincter
4. Serosa	Simple squamous epithelium (mesothelium) over a thin layer of loose connective tissue	The visceral peritoneum. Where an organ has no serosa, an **adventitia** of fibrous connective tissue covers it instead

The epithelium of the mucosa changes with the job of the region. The mouth, the oropharynx and laryngopharynx, the esophagus, and the anal canal are lined with **stratified squamous epithelium**, which stands up to rubbing by food. The stomach, small intestine, and large intestine are lined with **simple columnar epithelium**, which secretes and absorbs.

The outermost layer also changes. The serosa is present only where the tube lies in the abdominal cavity. Most of the esophagus, the part in the neck and thorax, has an adventitia instead. Only the short segment of esophagus below the diaphragm, just before the stomach, has a serosa. Retroperitoneal organs (Competency 112) have an adventitia on the side that faces the back wall.

Two places break the usual muscle pattern. The stomach has a third, oblique layer of muscle (Competency 115). The large intestine gathers its outer longitudinal layer into three bands, the teniae coli (Competency 117).

MEMORY TIP

Read the wall from the inside out: "Mucosa, Submucosa, Muscularis, Serosa." The adventitia replaces the serosa on the esophagus above the diaphragm.

COMPETENCY 110 LECTURE AND LAB

Enteric nerve plexuses

You should be able to: Locate the submucosal and myenteric plexuses in the gut wall and match each to the wall layer it sits within.

The wall of the alimentary canal has its own nerve supply, the **enteric nervous system**. It is made of two networks of neurons, called plexuses, that run the whole length of the tube inside the wall. Each plexus sits in a different layer, and the layer it sits in matches the job it controls.

The **submucosal plexus** (Meissner plexus) lies in the submucosa, just deep to the mucosa. It controls the glands of the mucosa and the blood flow to it. That fits its position right next to the lining.

The **myenteric plexus** (Auerbach plexus) lies inside the muscularis, between the inner circular layer and the outer longitudinal layer. It controls the contractions of those two muscle layers, which move food along the tube. That fits its position between the two muscle layers it runs.

Both plexuses connect to the autonomic nervous system. Parasympathetic fibers (mostly from the vagus nerve) and sympathetic fibers reach the gut wall and adjust what the plexuses do.

THE TWO ENTERIC PLEXUSES

Plexus	Wall layer it sits in	What it controls
Submucosal (Meissner) plexus	Submucosa	Secretion by the glands of the mucosa, and blood flow to the mucosa
Myenteric (Auerbach) plexus	Muscularis, between the inner circular and outer longitudinal layers	Contraction of the muscularis, which mixes food and moves it along

MEMORY TIP

The name tells you the layer. "Sub-mucosal" is in the submucosa. "Myenteric" comes from "myo", which means muscle, so it sits in the muscularis.

COMPETENCY 111 LECTURE AND LAB

Peritoneum and folds

You should be able to: Identify the parietal and visceral peritoneum, the peritoneal cavity, and the five folds greater omentum, lesser omentum, falciform ligament, mesentery, and mesocolon and state what each fold connects.

The **peritoneum** is the large serous membrane of the abdominopelvic cavity. Like the pleura around the lungs, it has two layers with a thin film of fluid between them. The fluid lets the organs slide against each other and against the body wall without friction.

THE LAYERS OF THE PERITONEUM AND THE SPACE BETWEEN THEM

Structure	Where it is
Parietal peritoneum	Lines the inner surface of the abdominal and pelvic walls
Visceral peritoneum	Covers the outer surface of the abdominal organs. On the organs of the canal it is the serosa
Peritoneal cavity	The thin space between the parietal and visceral layers. It holds only a small amount of serous fluid
Retroperitoneum	The region behind the parietal peritoneum, between it and the posterior abdominal wall (Competency 112)

The visceral peritoneum does more than cover organs. In many places two layers of peritoneum come together to form a **fold** (a double layer). These folds hold organs in place, attach them to the body wall or to each other, and carry blood vessels, lymphatic vessels, and nerves to the organs. Fat is stored between the two layers of many folds.

Five large folds are named in this course. When you learn each one, learn its two ends: what it hangs from and what it attaches to.

THE FIVE MAJOR PERITONEAL FOLDS

Fold	What it connects	Other features

Fold	What it connects	Other features
Greater omentum	Hangs from the greater curvature of the stomach (and the first part of the duodenum), drapes down in front of the small intestine, then folds back up and attaches to the transverse colon	The largest fold. A fatty apron that covers the intestines and holds many lymph nodes
Lesser omentum	Runs from the lesser curvature of the stomach and the first part of the duodenum up to the inferior surface of the liver	Its free edge carries the vessels and duct that serve the liver: the hepatic artery, the hepatic portal vein, and the common bile duct
Falciform ligament	Attaches the liver to the anterior abdominal wall and the inferior surface of the diaphragm	A thin, sickle-shaped fold. It marks the boundary between the right and left lobes of the liver on the anterior view. Its free lower edge holds the ligamentum teres (round ligament)
Mesentery	Attaches the jejunum and ileum to the posterior abdominal wall	Fan-shaped. Carries the blood vessels, lymphatic vessels, and nerves of the small intestine
Mesocolon	Attaches parts of the large intestine to the posterior abdominal wall: the transverse mesocolon holds the transverse colon, and the sigmoid mesocolon holds the sigmoid colon	Carries the vessels and nerves of those parts of the colon

In the cadaver, the greater omentum is the first thing you see when the abdominal wall is opened. Lift it upward, toward the head, to see the transverse colon on its back surface and the coils of small intestine below it.

MEMORY TIP

"Greater" goes with the greater curvature of the stomach and hangs low. "Lesser" goes with the lesser curvature and runs up to the liver. "Meso" means middle: the mesentery and mesocolon hold the intestines out from the back wall like a hammock.

COMPETENCY 112 LECTURE AND LAB

Retroperitoneal organs

You should be able to: Distinguish retroperitoneal organs from intraperitoneal ones and name the kidneys, pancreas, duodenum, and parts of the colon as retroperitoneal.

Abdominal organs are sorted by where they sit relative to the peritoneum.

An **intraperitoneal organ** is wrapped almost completely in visceral peritoneum. It hangs into the abdominal cavity on a fold such as a mesentery or an omentum. That fold lets the organ move. The stomach, the jejunum, the ileum, the transverse colon, the sigmoid colon, the liver, and the spleen are intraperitoneal.

A **retroperitoneal organ** lies behind the parietal peritoneum, against the posterior abdominal wall. This region behind the peritoneum is the **retroperitoneum**. Peritoneum covers only its front surface. It has no mesentery, so it is held firmly in place. Its back surface is covered by an adventitia, not a serosa.

The retroperitoneal organs in this course are the **kidneys**, the **pancreas**, the **duodenum**, and two parts of the colon, the **ascending colon** and the **descending colon**. The duodenum is retroperitoneal except for its short first part, just past the stomach. The pancreas is retroperitoneal except for the tip of its tail.

The colon alternates. The ascending colon is retroperitoneal, the transverse colon hangs on its mesocolon, the descending colon is retroperitoneal, and the sigmoid colon hangs on its mesocolon. So the two parts of the colon that

have a mesocolon are the two that are intraperitoneal.

RETROPERITONEAL AND INTRAPERITONEAL PARTS OF THE CANAL AND NEARBY ORGANS

Organ or part	Position	How you can tell
Stomach	Intraperitoneal	Hangs between the lesser and greater omenta
Duodenum	Retroperitoneal (except its first part)	Fixed to the posterior wall in a C-shape around the head of the pancreas
Jejunum and ileum	Intraperitoneal	Hang on the mesentery
Ascending colon	Retroperitoneal	Fixed to the right posterior wall
Transverse colon	Intraperitoneal	Hangs on the transverse mesocolon
Descending colon	Retroperitoneal	Fixed to the left posterior wall
Sigmoid colon	Intraperitoneal	Hangs on the sigmoid mesocolon
Pancreas	Retroperitoneal (except the tip of the tail)	Lies across the posterior wall behind the stomach
Kidneys	Retroperitoneal	Lie against the posterior wall on either side of the vertebral column

MEMORY TIP

SADPUCKER names the retroperitoneal structures: **S**uprarenal (adrenal) glands, **A**orta and inferior vena cava, **D**uodenum (except the first part), **P**ancreas (except the tail), **U**reters, **C**olon (ascending and descending), **K**idneys, **E**sophagus (the short part in the abdomen), **R**ectum.

COMPETENCY 113 LECTURE AND LAB

Mouth, pharynx, esophagus

You should be able to: Identify the oral cavity boundaries, hard and soft palate, uvula, fauces, tonsils, the three pharyngeal regions, and the esophagus with its hiatus.

The oral cavity

The **oral cavity** (the mouth) is the first part of the canal. Food is taken in here, chewed by the teeth, moved by the tongue, and mixed with saliva. It is lined with stratified squamous epithelium.

THE BOUNDARIES OF THE ORAL CAVITY

Boundary	What forms it
Anterior	The lips, around the opening of the mouth
Lateral (sides)	The cheeks, built around the buccinator muscle
Superior (roof)	The hard palate in front and the soft palate behind it
Inferior (floor)	The mylohyoid muscle. The tongue rests on the floor and fills most of it

Boundary	What forms it
Posterior	The fauces, the opening into the oropharynx

The **hard palate** is the bony front part of the roof. It is formed by the palatine processes of the maxillae and the horizontal plates of the palatine bones, covered by mucosa. It gives the tongue a firm surface to press food against.

The **soft palate** is the back part of the roof. It is made of skeletal muscle covered by mucosa, so it can move. The **uvula** is the finger-shaped projection that hangs down from the middle of its back edge. During swallowing, the soft palate and uvula rise together and close off the nasopharynx, so food does not enter the nasal cavity.

Two arches of mucosa curve down from the soft palate on each side: the **palatoglossal arch** in front and the **palatopharyngeal arch** behind it. The **fauces** is the opening between the oral cavity and the oropharynx, framed by these arches.

The tonsils

Tonsils are clusters of lymphatic tissue in the mucosa around the throat. They guard the entrance to the digestive and respiratory tracts.

THE TONSILS AROUND THE THROAT

Tonsil	Where it is
Palatine tonsils	Paired, in the lateral walls of the oropharynx, between the palatoglossal and palatopharyngeal arches. These are the tonsils you see when you look into an open mouth
Lingual tonsils	On the posterior part (the root) of the tongue, in the oropharynx
Pharyngeal tonsil	Single, on the posterior wall of the nasopharynx. Called the adenoids when enlarged

The pharynx

The **pharynx** (throat) is a funnel-shaped tube of skeletal muscle behind the nasal cavity, the oral cavity, and the larynx. Its muscles, the pharyngeal constrictors, squeeze food down during swallowing. It has three regions. Food passes through only the lower two.

THE THREE REGIONS OF THE PHARYNX

Region	Where it is	Carries	Lining
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar epithelium
Oropharynx	Behind the oral cavity, from the soft palate down to the epiglottis	Air and food	Stratified squamous epithelium
Laryngopharynx	Behind the larynx, from the epiglottis down to the start of the esophagus	Air and food	Stratified squamous epithelium

The esophagus and the esophageal hiatus

The **esophagus** is a muscular tube about 25 cm (10 inches) long. It begins at the lower end of the laryngopharynx, at the level of the cricoid cartilage. It runs down through the mediastinum, posterior to the trachea and anterior to the vertebral column. It stays collapsed (flat) except when food is passing through.

To reach the stomach, the esophagus passes through the diaphragm at the **esophageal hiatus**, an opening at about the level of the T10 vertebra. Below the diaphragm, a short abdominal segment joins the stomach at the cardia.

The muscularis of the esophagus changes along its length. The upper part is skeletal muscle, the middle part is a mix, and the lower part is smooth muscle. The lining is stratified squamous epithelium. The outer layer is an adventitia above the diaphragm and a serosa on the short segment below it.

COMPETENCY 114 LECTURE AND LAB

Major canal sphincters

You should be able to: Locate the upper and lower esophageal, pyloric, ileocecal, and internal and external anal sphincters and state which regions each one separates.

A **sphincter** is a ring of muscle around the tube that can squeeze it closed. Sphincters sit at the borders between regions. They control when contents pass from one region to the next, and most of them keep contents from flowing backward. Some are skeletal muscle. Others are smooth muscle and work without conscious control.

This course names six sphincters along the canal. Learn each one by the two regions on either side of it.

THE MAJOR SPHINCTERS OF THE ALIMENTARY CANAL, FROM TOP TO BOTTOM

Sphincter	Where it is	Regions it separates	Muscle type
Upper esophageal sphincter	At the top of the esophagus, at the level of the cricoid cartilage	Laryngopharynx from esophagus	Skeletal muscle
Lower esophageal sphincter (also called the cardiac sphincter or gastroesophageal sphincter)	At the bottom of the esophagus, where it joins the stomach just below the diaphragm	Esophagus from stomach (the cardia)	Smooth muscle
Pyloric sphincter	At the end of the stomach, around the pylorus	Stomach (pylorus) from duodenum	Smooth muscle (a thickening of the circular layer)
Ileocecal sphincter	At the end of the ileum, where it opens into the cecum	Small intestine (ileum) from large intestine (cecum)	Smooth muscle
Internal anal sphincter	Around the anal canal, inside the external sphincter	Anal canal from the outside of the body, at the anus	Smooth muscle (involuntary)
External anal sphincter	Around the anal canal, outside the internal sphincter	Anal canal from the outside of the body, at the anus	Skeletal muscle (voluntary)

The upper esophageal sphincter is skeletal muscle, continuous with the muscles of the pharynx. It stays closed between swallows, which keeps air out of the esophagus.

The lower esophageal sphincter is a zone of smooth muscle at the lower end of the esophagus. The diaphragm around the esophageal hiatus helps it stay closed. It keeps acidic stomach contents from flowing back up into the esophagus.

The pyloric sphincter is a thick ring of circular smooth muscle. It lets small amounts of stomach contents into the duodenum at a time. The ileocecal sphincter is a thickening of the smooth muscle at the end of the ileum. Together with

folds of mucosa, it forms the **ileocecal valve**, which keeps the contents of the large intestine from moving back into the ileum.

The anal canal has two sphincters, one inside the other. The internal anal sphincter is a thickening of the circular smooth muscle of the canal. The external anal sphincter is skeletal muscle, so it is the one you control voluntarily.

The **hepatopancreatic sphincter** (sphincter of Oddi) is on the lab list with the small intestine. It is not a sphincter of the canal itself. It controls the flow of bile and pancreatic juice into the duodenum at the major duodenal papilla, and it is covered with the accessory organs.

MEMORY TIP

Only the two ends of the canal have skeletal muscle sphincters: the upper esophageal sphincter at the top and the external anal sphincter at the bottom. Every sphincter in between is smooth muscle.

COMPETENCY 115 LECTURE AND LAB

Stomach regions and features

You should be able to: Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

The **stomach** is a J-shaped widening of the canal in the upper left abdomen, just below the diaphragm. It sits between the esophagus and the duodenum. It holds a meal, churns it, and mixes it with gastric juice, then releases it into the duodenum a little at a time. When empty it is small, and it stretches a great deal as it fills.

Regions

The stomach has four regions. Food enters at the cardia, collects in the body, and leaves through the pyloric part. The fundus is the dome that rises above the entrance.

THE REGIONS OF THE STOMACH

Region	Where it is
Cardia (cardiac region)	The small region around the opening from the esophagus
Fundus	The dome-shaped part that bulges above and to the left of the cardia, just under the diaphragm
Body	The large central part below the fundus
Pyloric part (pyloric region)	The funnel-shaped part that connects the body to the duodenum. It has three parts in order: the wide **pyloric antrum**, the narrow **pyloric canal**, and the **pylorus**, the outlet surrounded by the pyloric sphincter

Curvatures and inner folds

The stomach has two borders. The **lesser curvature** is the short, concave border on the right (medial) side, facing the liver. The **greater curvature** is the long, convex border on the left (lateral) side. The lesser omentum attaches along the lesser curvature, and the greater omentum hangs from the greater curvature.

Inside, the lining of an empty stomach is thrown into large folds called **rugae**. Rugae are folds of the mucosa and submucosa. They flatten out as the stomach fills, which lets it expand. The surface of the mucosa is dotted with **gastric pits**, which lead down into the gastric glands that make gastric juice.

The three-layered muscularis

The rest of the canal has two layers of smooth muscle in the muscularis. The stomach has three. The extra layer is the **oblique layer**, the innermost of the three. Its fibers run at an angle to the other two, so the stomach can squeeze and twist its contents as well as push them forward. This helps churn food into a liquid.

THE MUSCULARIS OF THE STOMACH, FROM THE LUMEN OUTWARD

Layer	Fiber direction
Inner oblique layer	Diagonal. This is the extra layer found only in the stomach
Middle circular layer	Around the stomach. Thickens at the pylorus to form the pyloric sphincter
Outer longitudinal layer	Along the length of the stomach

COMPETENCY 116 LECTURE AND LAB

Small intestine and surface area

You should be able to: Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

The **small intestine** runs from the pyloric sphincter to the ileocecal valve. It is the longest part of the canal, about 3 m (10 feet) in a living person, and it fills most of the middle and lower abdomen. Most chemical digestion and nearly all absorption of nutrients happen here. It has three regions.

THE REGIONS OF THE SMALL INTESTINE

Region	Length	Location and features
Duodenum	About 25 cm (10 inches); the shortest	A C-shaped curve that starts at the pyloric sphincter and wraps around the head of the pancreas. Mostly retroperitoneal. Bile and pancreatic juice enter it through the **major duodenal papilla** on its inner wall
Jejunum	About 1 m (3 feet)	The middle region, mostly in the upper left abdomen. Hangs on the mesentery
Ileum	About 2 m (6 feet); the longest	The last region, mostly in the lower right abdomen. Hangs on the mesentery. Ends at the ileocecal valve, where it joins the cecum. Its wall holds clusters of lymphatic tissue (Peyer patches)

There is no sharp border between the jejunum and the ileum. One gradually becomes the other along the mesentery.

Features that expand the surface area

Absorption happens across the surface of the lining, so more surface means more absorption. The small intestine increases its surface area at three levels, each one smaller than the last.

THREE LEVELS OF FOLDING IN THE SMALL INTESTINE, FROM LARGEST TO SMALLEST

Feature	What it is	Visible with
Circular folds (plicae circulares)	Permanent ridges of the mucosa and submucosa that run around the inside of the tube. Unlike the rugae of the stomach, they do not flatten when the intestine fills	The naked eye
Villi	Finger-like projections of the mucosa that cover the circular folds and give the lining a velvety look. Each villus has a core of lamina propria with a capillary network and a lacteal	A light microscope
Microvilli	Tiny projections of the plasma membrane on the free surface of each absorptive cell. Together they form the **brush border**	An electron microscope (they appear only as a fuzzy edge on a light microscope)

The **lacteal** is a lymphatic capillary in the center of each villus. Most absorbed fats are too large to enter the blood capillaries, so they enter the lacteal instead and travel in the lymph before reaching the blood.

MEMORY TIP

Fold, finger, fuzz. Circular folds are folds of the wall. Villi are fingers on the folds. Microvilli are the fuzz on each cell of each finger.

COMPETENCY 117 LECTURE AND LAB

Large intestine features

You should be able to: Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

The **large intestine** is the last part of the canal. It runs from the ileocecal valve to the anus, about 1.5 m (5 feet), and frames the small intestine on three sides. It is wider than the small intestine but much shorter. It absorbs water from what is left after digestion and forms and stores the feces. Its mucosa is simple columnar epithelium with many mucus-secreting goblet cells and no villi.

Regions

THE REGIONS OF THE LARGE INTESTINE, IN ORDER

Region	Where it is
Cecum	A blind pouch in the lower right abdomen, below the ileocecal valve. The first part of the large intestine
Vermiform appendix	A narrow, worm-shaped tube attached to the cecum. Its wall holds a large amount of lymphatic tissue
Ascending colon	Rises up the right side of the abdomen to the liver. Retroperitoneal
Hepatic flexure (right colic flexure)	The bend under the liver where the ascending colon turns left
Transverse colon	Crosses the abdomen from right to left, below the stomach. Hangs on the transverse mesocolon
Splenic flexure (left colic flexure)	The bend near the spleen where the transverse colon turns down

Region	Where it is
Descending colon	Runs down the left side of the abdomen. Retroperitoneal
Sigmoid colon	The S-shaped part in the pelvis that curves medially to join the rectum. Hangs on the sigmoid mesocolon
Rectum	The straight part in the pelvis, anterior to the sacrum and coccyx. Stores feces
Anal canal	The last few centimeters of the canal, passing through the pelvic floor and opening to the outside at the **anus**. Lined with stratified squamous epithelium near the anus. Surrounded by the internal and external anal sphincters

The **colon** has four parts: the ascending, transverse, descending, and sigmoid colon. The two flexures are the bends between them, not separate parts.

Features found only on the large intestine

Three surface features let you tell the large intestine from the small intestine at a glance, on a model or the cadaver.

THREE FEATURES UNIQUE TO THE LARGE INTESTINE

Feature	What it is
Teniae coli	Three flat ribbons of smooth muscle that run along the length of the cecum and colon. They are the outer longitudinal layer of the muscularis, gathered into three bands instead of forming a complete sheet
Haustra	The row of pouches that gives the colon its puckered look. They form because the teniae coli are shorter than the wall, so the wall bunches up between them
Omental appendices (epiploic appendages)	Small fat-filled pouches of visceral peritoneum that hang from the surface of the colon, mostly along the teniae coli

The teniae coli spread out into a complete layer again at the rectum, so the rectum has no haustra. The three teniae meet at the base of the appendix, which is one way to find the appendix on a specimen.

MEMORY TIP

Trace the colon as a frame around the small intestine: up the right side (ascending), a bend at the liver (hepatic flexure), across the top (transverse), a bend at the spleen (splenic flexure), down the left side (descending), then an S into the pelvis (sigmoid).

COMPETENCY 118 LECTURE AND LAB

Trace the pathway of food

You should be able to: Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

Food moves through the alimentary canal in one direction only. Every named segment below is one the food passes through, in order. The sphincters are listed at the borders where they sit.

1. **Oral cavity** food is taken in through the lips, chewed by the teeth, mixed with saliva, and shaped into a soft mass (a bolus) by the tongue
2. **Fauces** the tongue pushes the bolus back through the opening into the throat

3. **Oropharynx** the soft palate and uvula rise to close off the nasopharynx above
4. **Laryngopharynx** the epiglottis covers the larynx, so the bolus passes behind it toward the esophagus
5. **Upper esophageal sphincter** relaxes to let the bolus into the esophagus
6. **Esophagus** waves of muscle contraction carry the bolus down through the thorax and through the esophageal hiatus of the diaphragm
7. **Lower esophageal sphincter** relaxes to let the bolus into the stomach
8. **Cardia of the stomach** the entrance region of the stomach
9. **Body of the stomach** food is churned and mixed with gastric juice until it becomes a liquid called chyme
10. **Pyloric antrum, pyloric canal, and pylorus** the funnel-shaped exit region of the stomach
11. **Pyloric sphincter** lets small amounts of chyme into the small intestine
12. **Duodenum** bile and pancreatic juice enter at the major duodenal papilla
13. **Jejunum** digestion continues and most nutrients are absorbed
14. **Ileum** absorption continues
15. **Ileocecal sphincter (ileocecal valve)** lets the remaining material into the large intestine
16. **Cecum** the pouch at the start of the large intestine
17. **Ascending colon** up the right side of the abdomen
18. **Hepatic (right colic) flexure** the bend under the liver
19. **Transverse colon** across the abdomen from right to left
20. **Splenic (left colic) flexure** the bend near the spleen
21. **Descending colon** down the left side of the abdomen
22. **Sigmoid colon** the S-shaped curve into the pelvis
23. **Rectum** feces are stored here
24. **Anal canal** passes through the internal and external anal sphincters
25. **Anus** feces leave the body

Two named structures are off the main path. The fundus of the stomach bulges above the cardia, so food collects there but does not have to pass through it to move on. The vermiform appendix is a blind tube hanging from the cecum, so food never travels through it.

The nasopharynx is also not on the path. It carries only air, and the soft palate seals it off during swallowing.

MEMORY TIP

Group the path into the six organs of the canal first: mouth, pharynx, esophagus, stomach, small intestine, large intestine. Then fill in the named parts inside each organ, and put a sphincter at each border.

COMPETENCY 119 LECTURE

Clinical anatomy of canal disorders

You should be able to: Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

Each disorder of the alimentary canal starts in a specific structure. If you know where the structure is and what layers it has, you can predict where the patient feels it and how serious it can become.

DISORDERS OF THE ALIMENTARY CANAL AND THE STRUCTURES THEY AFFECT

Disorder	Structure affected	What changes in that structure
Gastroesophageal reflux disease (GERD)	Lower esophageal sphincter and the lower esophagus	The sphincter does not close well, so acidic stomach contents flow back into the esophagus. The stratified squamous lining of the esophagus is not protected against acid, so it becomes irritated and inflamed
Hiatal hernia	Esophageal hiatus of the diaphragm and the upper stomach	Part of the stomach pushes up through the esophageal hiatus into the thorax. This weakens the support the diaphragm gives the lower esophageal sphincter and often comes with reflux
Peptic ulcer	Mucosa of the stomach or the duodenum	An open sore where the mucosa has been worn away. It can deepen through the submucosa and muscularis. Common causes are infection with the bacterium Helicobacter pylori and regular use of NSAID pain relievers
Appendicitis	Vermiform appendix	The lumen of the appendix becomes blocked, and the appendix swells and becomes inflamed. If it bursts, its contents spill into the peritoneal cavity and inflame the peritoneum (peritonitis)
Diverticulosis	Wall of the colon, most often the sigmoid colon	Small pouches (diverticula) of mucosa and submucosa bulge outward through weak spots in the muscularis. When the pouches become inflamed, the condition is called diverticulitis
Inflammatory bowel disease (IBD)	Wall of the small and large intestine	Long-term inflammation of the wall. **Crohn disease** can affect any part of the canal from mouth to anus, most often the ileum and colon, and can pass through every layer of the wall. **Ulcerative colitis** affects only the colon and rectum and stays mostly in the mucosa
Hemorrhoids	Veins in the wall of the anal canal and lower rectum	The veins swell and stretch, forming soft bulges in the wall of the canal or around the anus
Colorectal cancer	Mucosa of the colon or rectum	Cells of the lining grow out of control, usually starting as a small growth (a polyp). The tumor can then grow through the deeper layers of the wall

The disorders that affect the wall of a region, such as ulcers, diverticula, IBD, and cancer, become more serious as they reach deeper layers of the wall.

WHERE A PROBLEM SITS PREDICTS THE SYMPTOM

Each region of the canal has its own position in the body, so where a problem sits predicts where the patient feels it. A weak **lower esophageal sphincter** brings a burning feeling behind the sternum, called heartburn, because the esophagus lies in the thorax just behind it. An inflamed **appendix** usually brings pain to the lower right abdomen, where the cecum and appendix lie.

The four-layer wall explains how a problem gets worse. A **peptic ulcer** begins in the mucosa. As it erodes deeper, it passes through the submucosa, where it can open a blood vessel and bleed, and then through the muscularis. If it erodes through the serosa, the wall is perforated and the contents of the lumen spill into the peritoneal cavity.

CHAPTER 4

The Accessory Digestive Organs

DIGESTIVE

COMPETENCIES 120 TO 125

Covers the teeth and tongue, the salivary glands, the liver, the gallbladder and the path of bile, and the pancreas. The focus is on the structures and the job each one does.

COMPETENCIES

1. **120** Teeth and tongue
2. **121** Salivary glands
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WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Accessory digestive organs

drsrennie-stack.github.io/new-build-bio4-solano/accessory-digestive-concept-videos.html

COMPETENCY 120 LECTURE AND LAB

Teeth and tongue

You should be able to: Identify the teeth and tongue with the lingual frenulum, papillae, intrinsic and extrinsic muscles, and lingual glands.

The teeth and the tongue are the two accessory organs that touch food directly. The teeth cut and grind food. The tongue moves food around the mouth, presses it against the teeth, shapes it into a soft ball called a **bolus**, and pushes the bolus back toward the pharynx for swallowing. The tongue also shapes sounds for speech.

The teeth

Each tooth sits in a socket called an **alveolus** in the maxilla or the mandible. A thin **periodontal ligament** holds the tooth in its socket. This tooth and socket joint is a fibrous joint called a **gomphosis**. Children have 20 **deciduous teeth** (baby teeth). Adults have 32 **permanent teeth**.

THE FOUR TYPES OF TEETH (ADULT NUMBERS)

Tooth type	Number in adults	Shape and job
Incisors	8	Front teeth with a flat, chisel-like edge. They cut food.
Canines (cuspids)	4	One pointed cusp. They tear and pierce food.
Premolars (bicuspids)	8	Two cusps. They crush and grind food.
Molars	12, including the four third molars (wisdom teeth)	Several cusps on a broad surface. They do the heaviest grinding.

Every tooth has the same basic parts. The **crown** is the part above the gum, or **gingiva**. The **neck** sits at the gum line. The **root** is held in the socket.

- **Enamel** covers the crown. It is the hardest substance in the body.
- **Dentin** forms most of the tooth, under the enamel.

- The **pulp cavity** in the center holds the pulp: blood vessels, nerves, and connective tissue. It continues down each root as the **root canal**.
- **Cementum** covers the root and anchors it to the periodontal ligament.

The tongue

The **tongue** is a mass of skeletal muscle covered by a mucous membrane. It occupies the floor of the mouth. The muscle under it, the **mylohyoid**, forms the actual floor. Two groups of skeletal muscle make up the tongue, and the **hypoglossal nerve** (cranial nerve XII) supplies all of them except the palatoglossus, which the vagus nerve supplies.

The **extrinsic tongue muscles** start on bone or the soft palate outside the tongue and end inside it. They move the whole tongue: out, back, up, down, and side to side. The **intrinsic tongue muscles** start and end inside the tongue. Their fibers run lengthwise, across, and up and down. They change the shape of the tongue, letting it curl, flatten, narrow, and shorten for speech and swallowing.

THE EXTRINSIC TONGUE MUSCLES

Muscle	Starts on	Main action
Genioglossus	The inside of the mandible at the chin	Pushes the tongue out (protrusion) and lowers it
Hyoglossus	The hyoid bone	Pulls the tongue down and back
Styloglossus	The styloid process of the temporal bone	Pulls the tongue up and back
Palatoglossus	The soft palate	Raises the back of the tongue

The **lingual frenulum** is a midline fold of mucous membrane on the underside of the tongue. It anchors the tongue to the floor of the mouth and limits how far the tongue can move back. A frenulum that is too short, called **ankyloglossia** (tongue-tie), restricts tongue movement and can affect speech.

The top surface of the tongue is covered with small bumps called **lingual papillae**. There are four kinds. Three of them carry taste buds. The most common kind, the filiform papillae, has no taste buds and only gives the tongue a rough surface that grips food.

THE FOUR KINDS OF LINGUAL PAPILLAE

Papillae	Where and what they look like	Taste buds?
Filiform papillae	Thin, pointed, and the most numerous. They cover most of the top of the tongue.	No. They give the tongue grip.
Fungiform papillae	Mushroom-shaped. Scattered among the filiform papillae, most on the tip and sides.	Yes
Vallate (circumvallate) papillae	Large, round, and few (about 8 to 12) in a V-shaped row near the back of the tongue. A trench surrounds each one.	Yes
Foliate papillae	Leaf-like folds on the sides of the tongue toward the back.	Yes

The **lingual glands** are small glands in the connective tissue of the tongue. They release mucus and a watery fluid that contains **lingual lipase**, an enzyme that starts the breakdown of fats. They are among the minor salivary glands.

MEMORY TIP

Filiform papillae are for friction. They are the one kind of papilla with no taste buds.

COMPETENCY 121 LECTURE AND LAB

Salivary glands

You should be able to: Locate the parotid, submandibular, and sublingual glands and state where the duct of each opens.

Saliva moistens and cleans the mouth, lubricates food so it can be swallowed, and starts the chemical breakdown of starch. Small **minor salivary glands** in the lining of the lips, cheeks, palate, and tongue (the labial, buccal, palatal, and lingual glands) keep the mouth moist all the time. Most saliva, though, comes from three pairs of large glands that sit outside the lining of the mouth. Each one sends its saliva into the mouth through a duct.

The **parotid gland** is the largest. It lies under the skin in front of and below the ear, over the masseter muscle. Its duct, the **parotid duct**, runs forward across the masseter, passes through the buccinator muscle of the cheek, and opens on the inside of the cheek opposite the upper second molar. The parotid makes a watery (serous) saliva.

The **submandibular gland** lies along the inner surface of the body of the mandible, in the floor of the mouth. Its duct, the **submandibular duct**, runs forward under the lining of the floor of the mouth and opens on a small bump on each side of the lingual frenulum. It makes a mix of watery and mucous saliva.

The **sublingual gland** is the smallest of the three. It lies under the tongue, in the floor of the mouth just under the lining, above the submandibular gland. It drains through many small ducts that open along the floor of the mouth under the tongue. Its saliva is mostly mucous.

THE THREE LARGE PAIRS OF SALIVARY GLANDS

Gland	Location	Where the duct opens	Type of saliva
Parotid gland	In front of and below the ear, between the skin and the masseter muscle	Inside the cheek, opposite the upper second molar	Watery (serous)
Submandibular gland	Floor of the mouth, along the inner surface of the mandible	On each side of the lingual frenulum	Mixed watery and mucous
Sublingual gland	Under the tongue in the floor of the mouth, above the submandibular gland	Many small ducts opening into the floor of the mouth	Mostly mucous

MEMORY TIP

Each name tells you the spot. Parotid means "beside the ear." Submandibular means "below the mandible." Sublingual means "below the tongue."

COMPETENCY 122 LECTURE AND LAB

Liver gross and lobule anatomy

You should be able to: Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

The **liver** is the largest gland in the body. It lies just under the diaphragm, mostly in the upper right part of the abdomen, and reaches across the midline to the left. The lower ribs protect it. Its smooth upper surface curves against the diaphragm. Its lower surface, the **visceral surface**, faces the stomach, duodenum, gallbladder, right kidney, and colon. The liver makes bile and processes the blood that arrives from the digestive tract.

Lobes and ligaments

The liver has four lobes. From the front you see only two. The **right lobe** is the largest. The smaller **left lobe** reaches over the stomach. The **falciform ligament** marks the line between them on the front of the liver. The other two lobes are small

and are seen from below and behind. The **quadrate lobe** is on the inferior surface, between the gallbladder and the ligamentum teres. The **caudate lobe** is on the posterior surface, beside the inferior vena cava.

Between the quadrate and caudate lobes is the **porta hepatis**, the doorway of the liver. The hepatic portal vein and the hepatic artery proper enter here, and the right and left hepatic ducts leave here.

The falciform ligament is a sickle-shaped fold of peritoneum. It attaches the liver to the diaphragm and the anterior abdominal wall. In its free lower edge runs the **ligamentum teres**, also called the round ligament. It is a fibrous cord that runs from the umbilicus to the liver. The ligamentum teres is what remains of the fetal **umbilical vein**, which carried oxygen-rich blood from the placenta to the fetus. The **lesser omentum** is a double fold of peritoneum that runs from the lesser curvature of the stomach and the start of the duodenum to the visceral surface of the liver. Its free right edge carries the hepatic portal vein, the hepatic artery proper, and the common bile duct.

THE LOBES AND LIGAMENTS OF THE LIVER

Structure	Where to find it	What it is
Right lobe	Right of the falciform ligament	The largest lobe
Left lobe	Left of the falciform ligament, over the stomach	The second largest lobe
Quadrate lobe	Inferior surface, between the gallbladder and the ligamentum teres	A small, square-shaped lobe
Caudate lobe	Posterior surface, beside the inferior vena cava	A small lobe with a tail-like process
Falciform ligament	Anterior surface, from the liver to the diaphragm and anterior abdominal wall	A fold of peritoneum that separates the right and left lobes
Ligamentum teres (round ligament)	In the free lower edge of the falciform ligament	A fibrous cord; the remnant of the fetal umbilical vein
Lesser omentum	From the lesser curvature of the stomach to the liver	A peritoneal fold; its free edge carries the portal vein, hepatic artery proper, and common bile duct

The liver lobule

Liver tissue is built from thousands of small units called **liver lobules**. In a slide, a lobule is roughly six-sided. A **central vein** runs through its center. Plates of liver cells, called **hepatic laminae**, radiate out from the central vein toward the edges, like the spokes of a wheel.

The **hepatocytes** are the liver cells in those plates. They do most of the liver's work, including making bile. Between the plates run the **hepatic sinusoids**, wide and leaky capillaries. Blood in the sinusoids flows past the hepatocytes from the edge of the lobule toward the central vein. Phagocytic cells called **stellate macrophages** (Kupffer cells) line the sinusoids and remove worn-out blood cells and bacteria.

The **bile canaliculi** are tiny channels between neighboring hepatocytes inside each plate. They collect the bile the hepatocytes make and carry it outward to the edge of the lobule, the opposite direction from the blood. At each corner of the lobule sits a **portal triad**. It has three parts: a branch of the **hepatic portal vein**, a branch of the **hepatic artery**, and a **bile duct**. The two vessels bring blood into the lobule. The bile duct carries bile away.

THE PARTS OF A LIVER LOBULE

Structure	Where it is	What it does
Central vein	Center of the lobule	Collects blood from the sinusoids
Hepatocytes	In plates (hepatic laminae) that radiate from the central vein	The working liver cells; they make bile

Structure	Where it is	What it does
Hepatic sinusoids	Between the plates of hepatocytes	Leaky capillaries that carry mixed blood toward the central vein
Bile canaliculi	Between neighboring hepatocytes, inside each plate	Collect bile and carry it toward the portal triad
Portal triad	At each corner of the lobule	A branch of the hepatic portal vein, a branch of the hepatic artery, and a bile duct

Blood flows through the liver in this order:

1. **Hepatic portal vein and hepatic artery proper** enter the liver at the porta hepatis. The portal vein brings nutrient-rich, oxygen-poor blood from the digestive tract and spleen. The hepatic artery brings oxygen-rich blood from the aorta.
2. **Portal triad branches** small branches of both vessels run to the corners of every lobule.
3. **Hepatic sinusoids** blood from the two vessels mixes and flows toward the center of the lobule, past the hepatocytes.
4. **Central vein** collects the blood at the center of each lobule.
5. **Hepatic veins** the central veins join into larger veins that leave the back of the liver.
6. **Inferior vena cava** receives the hepatic veins just below the diaphragm and carries the blood to the right atrium.

MEMORY TIP

In a lobule, blood flows in toward the central vein and bile flows out toward the portal triad. They travel in opposite directions.

COMPETENCY 123 LECTURE AND LAB

Gallbladder and bile pathway

You should be able to: Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

The **gallbladder** is a pear-shaped muscular sac about 8 to 10 cm long. It sits in a shallow hollow on the visceral surface of the right lobe of the liver, the surface that faces down and back. It stores bile between meals and concentrates it by removing water. When fatty food enters the duodenum, the smooth muscle in the gallbladder wall contracts and squeezes bile out.

The gallbladder has three regions. The **fundus** is the wide, rounded end. It sticks out past the lower edge of the liver toward the front. The **body** is the main middle part. The **neck** is the narrow end that points toward the porta hepatis and turns into the cystic duct.

The ducts that carry bile are called the **biliary tree**. The **right hepatic duct** and **left hepatic duct** drain bile from the two sides of the liver and leave at the porta hepatis. They join to form the **common hepatic duct**. The **cystic duct** runs from the neck of the gallbladder and joins the common hepatic duct. Together they form the **common bile duct**. The common bile duct runs down in the edge of the lesser omentum, passes behind the first part of the duodenum, and runs through the head of the pancreas.

In the wall of the duodenum, the common bile duct joins the main pancreatic duct at a short, widened chamber called the **hepatopancreatic ampulla**. The ampulla opens into the descending (second) part of the duodenum at a small bump called the **major duodenal papilla**. A ring of smooth muscle, the **sphincter of Oddi** (hepatopancreatic sphincter), surrounds the ampulla and controls the flow.

The cystic duct carries bile in both directions. When the sphincter of Oddi is closed, bile backs up the common bile duct and flows through the cystic duct into the gallbladder for storage. When the gallbladder contracts, bile leaves through the same cystic duct and runs down the common bile duct to the duodenum.

THE GALLBLADDER AND THE BILE DUCTS

Structure	What it is
Fundus of the gallbladder	The wide, rounded end; it sticks out past the lower edge of the liver
Body of the gallbladder	The main middle part
Neck of the gallbladder	The narrow end that continues as the cystic duct
Right and left hepatic ducts	Drain bile from the right and left sides of the liver
Common hepatic duct	Formed where the right and left hepatic ducts join
Cystic duct	Carries bile into and out of the gallbladder
Common bile duct	Formed where the common hepatic duct and the cystic duct join
Hepatopancreatic ampulla	The ampulla of Vater; the chamber where the common bile duct and main pancreatic duct meet in the duodenal wall
Major duodenal papilla	The bump in the duodenal lining where the ampulla opens
Sphincter of Oddi	The smooth muscle ring around the ampulla that controls flow into the duodenum

Bile flows from the liver to the duodenum in this order:

1. **Hepatocytes** make the bile.
2. **Bile canaliculi** tiny channels between hepatocytes collect it and carry it to the edge of the lobule.
3. **Bile ducts of the portal triads** carry bile out of each lobule and join into larger ducts.
4. **Right and left hepatic ducts** drain the two sides of the liver and leave at the porta hepatis.
5. **Common hepatic duct** formed by the right and left hepatic ducts.
6. **Cystic duct and gallbladder** bile can turn up the cystic duct into the gallbladder for storage, then come back down the cystic duct when the gallbladder contracts.
7. **Common bile duct** formed where the common hepatic duct and the cystic duct join.
8. **Hepatopancreatic ampulla** the common bile duct joins the main pancreatic duct.
9. **Sphincter of Oddi and major duodenal papilla** the sphincter opens and bile enters through the papilla.
10. **Duodenum** bile mixes with food in the small intestine.

MEMORY TIP

The duct names tell you where each duct comes from. Hepatic means liver. Cystic means bladder, here the gallbladder. The common bile duct is where both come together.

COMPETENCY 124 LECTURE AND LAB

Pancreas structure

You should be able to: Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

The **pancreas** is a long, soft gland that lies across the posterior abdominal wall, behind the stomach. It is retroperitoneal, which means it sits behind the peritoneum. The pancreas is both an exocrine gland, sending pancreatic juice through ducts into the duodenum, and an endocrine gland, releasing hormones into the blood.

It has three regions. The **head of the pancreas** is the wide end. It sits in the C-shaped curve of the duodenum, to the right of the midline. The **body of the pancreas** crosses the midline behind the stomach, in front of the aorta and the vertebral column. The **tail of the pancreas** tapers toward the left and ends near the spleen.

The **main pancreatic duct** (duct of Wirsung) runs the whole length of the gland, from the tail through the body to the head, collecting pancreatic juice along the way. In the wall of the duodenum it joins the common bile duct at the hepatopancreatic ampulla. The ampulla opens at the major duodenal papilla, and the sphincter of Oddi controls the flow. The smaller **accessory pancreatic duct** (duct of Santorini) drains part of the head. It opens on its own into the duodenum at the **minor duodenal papilla**, a little above the major papilla.

Under the microscope the pancreas has two kinds of tissue. Most of it is made of **acini**, small clusters of gland cells around tiny ducts. The acini are the exocrine part and make pancreatic juice, which carries digestive enzymes. Scattered among the acini are the **pancreatic islets** (islets of Langerhans). They are the endocrine part and release hormones into the blood. On a stained slide the islets look like pale, round patches surrounded by darker acini.

THE REGIONS, DUCTS, AND TISSUE OF THE PANCREAS

Structure	What it is
Head of the pancreas	The wide end, in the C-shaped curve of the duodenum
Body of the pancreas	The middle part, crossing the midline behind the stomach
Tail of the pancreas	The narrow left end, near the spleen
Main pancreatic duct	The duct of Wirsung; runs the length of the gland and joins the common bile duct at the hepatopancreatic ampulla
Accessory pancreatic duct	The duct of Santorini; drains part of the head and opens separately at the minor duodenal papilla
Major duodenal papilla	Where the hepatopancreatic ampulla delivers bile and pancreatic juice into the duodenum
Sphincter of Oddi	The smooth muscle ring around the ampulla that controls flow into the duodenum
Acini	The exocrine clusters that make pancreatic juice; about 99 percent of the gland
Pancreatic islets	The endocrine clusters that release hormones into the blood; about 1 percent of the gland

MEMORY TIP

Match the size words. The main pancreatic duct opens at the major papilla. The accessory pancreatic duct opens at the minor papilla.

COMPETENCY 125 LECTURE

Clinical anatomy of accessory disorders

You should be able to: Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

Each of these disorders affects one accessory organ or its ducts. Learn the organ first, then what changes in it.

DISORDERS OF THE ACCESSORY ORGANS

Disorder	Organ affected	What changes in the structure
Gallstones	Gallbladder and bile ducts	Hard deposits form from bile inside the gallbladder. A stone can move into the cystic duct or the common bile duct and block it.
Cholecystitis	Gallbladder	The gallbladder wall becomes inflamed and swollen, most often because a gallstone blocks the cystic duct and traps bile inside.
Hepatitis	Liver	The liver is inflamed and its hepatocytes are damaged, most often by a virus and also by alcohol or some drugs.
Cirrhosis	Liver	After long-term damage, hepatocytes are replaced by bands of scar tissue and lumps of regrowing cells. The lobules lose their normal pattern, and blood has trouble flowing through the liver.
Jaundice	Liver or bile ducts (seen in the skin and eyes)	Bilirubin, a yellow pigment from the breakdown of old red blood cells, builds up in the blood and stains the skin and the whites of the eyes yellow. It happens when the liver cannot clear bilirubin into bile, when a duct is blocked, or when red blood cells break down too fast.
Pancreatitis	Pancreas	The pancreas becomes inflamed and swollen. Digestive enzymes become active inside the gland and damage its own tissue. Gallstones and alcohol are common causes.
Mumps	Parotid glands	A viral infection makes the parotid glands swell, so the face puffs out in front of and below the ears.

A stone at the ampulla causes two problems at once because two ducts share that one opening. The common bile duct and the main pancreatic duct join at the hepatopancreatic ampulla, which empties into the duodenum through the major duodenal papilla. A gallstone that travels down the common bile duct and lodges in the ampulla plugs that shared exit. Bile backs up into the liver and the blood, which can cause jaundice. Pancreatic juice backs up into the pancreas, where its enzymes can damage the gland and cause pancreatitis.

SHARED DUCTS, SHARED PROBLEMS

The common bile duct and the main pancreatic duct meet at the **hepatopancreatic ampulla** before they reach the duodenum. Because they share that last passage, a gallstone stuck at the ampulla can block bile and pancreatic juice at the same time. This is one way gallstones lead to pancreatitis.

The liver gets blood from two sources: nutrient-rich blood from the **hepatic portal vein** and oxygen-rich blood from the **hepatic artery**. Blood from the stomach and intestines passes through the liver before it goes anywhere else. That is why cancers of the digestive tract often spread to the liver.

CHAPTER 5

Vessels and Nerves of the Abdomen and Lower Limb

LAB ONLY

COMPETENCIES 126 TO 127

A reference for the lab. These competencies are tested on the lab practical: find each vessel and nerve on the models and the cadaver, and say where it runs and what it serves.

COMPETENCIES

1. **126** Abdominal aorta and portal system
2. **127** Lower limb vessels and nerves

COMPETENCY 126 LAB

Abdominal aorta and portal system

You should be able to: Identify the major branches of the abdominal aorta and the vessels of the hepatic portal system on the cadaver.

The **abdominal aorta** is the part of the aorta below the diaphragm. It enters the abdomen through the aortic hiatus of the diaphragm at about the level of T12. It runs down the front of the lumbar vertebrae, just left of the midline, with the inferior vena cava on its right. At about L4 it divides into the right and left **common iliac arteries**.

Its branches come in two groups. The unpaired branches leave the front of the aorta and supply the digestive organs. The paired branches leave the sides and supply the kidneys, the gonads, and other structures on the right and left.

The three unpaired branches, from top to bottom

The **celiac trunk** leaves the aorta just below the diaphragm. It is short and splits almost at once into three arteries. The **left gastric artery** supplies the stomach along its lesser curvature and the lower end of the esophagus. The **splenic artery** runs to the left along the top of the pancreas to the spleen and gives branches to the pancreas and stomach on the way. The **common hepatic artery** runs to the right. It gives off the **hepatic artery proper**, which supplies the liver and gallbladder, along with branches to the stomach, duodenum, and pancreas.

The **superior mesenteric artery** leaves the aorta just below the celiac trunk. It runs in the mesentery and supplies most of the small intestine, part of the pancreas, and the large intestine from the cecum through most of the transverse colon. The **inferior mesenteric artery** leaves lower down. It supplies the rest of the large intestine: the last part of the transverse colon, the descending colon, the sigmoid colon, and the upper rectum.

THE BRANCHES OF THE ABDOMINAL AORTA ON YOUR LIST, FROM TOP TO BOTTOM

Branch	About where it leaves	What it supplies
Celiac trunk (unpaired)	T12, just below the diaphragm	Splits into the left gastric, splenic, and common hepatic arteries
Left gastric artery	From the celiac trunk	The stomach (lesser curvature) and the lower esophagus
Splenic artery	From the celiac trunk	The spleen, with branches to the pancreas and stomach
Common hepatic artery	From the celiac trunk	The liver and gallbladder, with branches to the stomach, duodenum, and pancreas
Superior mesenteric artery (unpaired)	L1	Most of the small intestine, part of the pancreas, the cecum, appendix, ascending colon, and most of the transverse colon

Branch	About where it leaves	What it supplies
Renal arteries (paired)	L1 to L2	The kidneys
Gonadal arteries (paired)	L2	The testes (testicular arteries) or the ovaries (ovarian arteries)
Inferior mesenteric artery (unpaired)	L3	The last part of the transverse colon, the descending colon, the sigmoid colon, and the upper rectum
Right and left common iliac arteries	L4, where the aorta ends	The pelvis and the lower limbs, through the internal and external iliac arteries

MEMORY TIP

The three gut branches go in the same order as the gut. The celiac trunk feeds the top (stomach, liver, spleen, pancreas, start of the duodenum). The superior mesenteric feeds the middle (small intestine and the colon up through most of the transverse colon). The inferior mesenteric feeds the end (the rest of the colon and the upper rectum).

Arteries of the pelvis

Each common iliac artery divides in front of the sacroiliac joint into two branches. The **internal iliac artery** dives into the pelvis. It supplies the urinary bladder, the rectum, the reproductive organs, the walls of the pelvis, the gluteal muscles, and the external genitalia. The **external iliac artery** runs along the psoas major toward the thigh. It passes under the inguinal ligament and becomes the femoral artery.

The **medial umbilical ligament** is a fibrous cord on the inside of the anterior abdominal wall that runs up from the pelvis toward the umbilicus. It is what remains of the fetal **umbilical artery**, a branch of the internal iliac artery that carried blood to the placenta. The part of the umbilical artery closest to the internal iliac stays open after birth and supplies the bladder.

Veins of the abdomen and pelvis

The **inferior vena cava** (IVC) is formed at about L5, where the right and left **common iliac veins** join. It runs up the right side of the aorta, passes through the diaphragm at about T8, and empties into the right atrium. Each common iliac vein is formed by an **internal iliac vein**, which drains the pelvis, and an **external iliac vein**, which drains the lower limb. The external iliac vein is the continuation of the femoral vein above the inguinal ligament.

The **renal veins** drain the kidneys into the IVC. The left renal vein is longer because it crosses in front of the aorta to reach the IVC. The **gonadal veins** drain the testes or ovaries. The right gonadal vein empties straight into the IVC, while the left gonadal vein empties into the left renal vein. The **hepatic veins** drain the liver into the IVC just below the diaphragm. The stomach, intestines, spleen, and pancreas do not drain straight into the IVC. Their blood goes to the liver first, through the hepatic portal system.

VEINS OF THE ABDOMEN AND PELVIS

Vein	Drains	Empties into
External iliac vein	The lower limb (continues from the femoral vein)	Common iliac vein
Internal iliac vein	The pelvis	Common iliac vein
Right and left common iliac veins	The pelvis and lower limb	Join to form the inferior vena cava

Vein	Drains	Empties into
Renal veins	The kidneys	Inferior vena cava
Right gonadal vein	The right testis or ovary	Inferior vena cava
Left gonadal vein	The left testis or ovary	Left renal vein
Hepatic veins	The liver	Inferior vena cava, just below the diaphragm
Inferior vena cava	Most of the body below the diaphragm	Right atrium

The hepatic portal system

A **portal system** is a route in which blood passes through two capillary beds in a row before it returns to the heart. In the **hepatic portal system**, the first capillary bed is in the walls of the stomach, intestines, spleen, and pancreas. The second capillary bed is the hepatic sinusoids of the liver. The **hepatic portal vein** connects the two. This route sends nutrient-rich blood from the gut to the liver to be processed before it reaches the rest of the body.

The hepatic portal vein forms behind the neck of the pancreas, where two veins join. The **superior mesenteric vein** runs beside the superior mesenteric artery and drains the small intestine and the colon up through most of the transverse colon. The **splenic vein** runs from the spleen along the back of the pancreas and collects blood from the spleen, pancreas, and part of the stomach. The **inferior mesenteric vein** drains the descending colon, sigmoid colon, and upper rectum, and usually empties into the splenic vein. The portal vein then runs up in the free edge of the lesser omentum and enters the liver at the porta hepatis.

THE VESSELS OF THE HEPATIC PORTAL SYSTEM

Vessel	Drains	Empties into
Superior mesenteric vein	The small intestine, cecum, ascending colon, and most of the transverse colon	Joins the splenic vein to form the hepatic portal vein
Inferior mesenteric vein	The descending colon, sigmoid colon, and upper rectum	Usually the splenic vein
Splenic vein	The spleen, pancreas, and part of the stomach	Joins the superior mesenteric vein to form the hepatic portal vein
Hepatic portal vein	All of the above	The liver, at the porta hepatis, then the hepatic sinusoids
Hepatic veins	The liver (after the sinusoids and central veins)	Inferior vena cava

A glucose molecule absorbed by the small intestine travels to the heart in this order:

1. **Capillaries of an intestinal villus** the glucose enters the blood here. This is the first capillary bed.
2. **Small veins of the intestinal wall and mesentery** carry the blood away from the intestine.
3. **Superior mesenteric vein** collects blood from the small intestine.
4. **Hepatic portal vein** formed where the superior mesenteric and splenic veins join; carries the blood to the liver.
5. **Hepatic sinusoids** the second capillary bed, where the blood passes the hepatocytes.
6. **Central vein** collects blood at the center of each lobule.
7. **Hepatic veins** carry the blood out of the liver.
8. **Inferior vena cava** receives the hepatic veins just below the diaphragm.

9. **Right atrium** the blood returns to the heart.

COMPETENCY 127 LAB

Lower limb vessels and nerves

You should be able to: Identify the femoral and lower-limb blood vessels and the major nerves of the abdomen and lower limb on the cadaver.

In the lower limb, arteries, veins, and nerves run together in bundles between the muscles you already know. Use those muscles as landmarks to find them. When you name a vessel, say right or left and artery or vein.

Arteries of the lower limb

One continuous artery runs from the abdomen to the foot and changes its name at set landmarks.

1. **External iliac artery** runs along the pelvic brim from the common iliac artery.
2. **Femoral artery** the external iliac becomes the femoral artery as it passes under the inguinal ligament.
3. **Popliteal artery** the femoral artery becomes the popliteal artery after it passes through an opening in the adductor magnus (the adductor hiatus) to reach the back of the knee.
4. **Anterior and posterior tibial arteries** the popliteal artery divides just below the knee.
5. **Dorsalis pedis and plantar arteries** the anterior tibial becomes the dorsalis pedis artery on the top of the foot; the posterior tibial divides into plantar arteries in the sole.

The **femoral artery** enters the thigh under the inguinal ligament, about halfway between the anterior superior iliac spine and the pubic symphysis. You can feel its pulse there. It first runs through the **femoral triangle**, a space bounded by the inguinal ligament above, the sartorius on the lateral side, and the adductor longus on the medial side. In the triangle the femoral vein lies on the medial side of the artery and the femoral nerve lies on the lateral side. A large branch, the **deep femoral artery**, leaves here and supplies most of the thigh muscles. The femoral artery then runs down under the sartorius and through the adductor hiatus to become the **popliteal artery** behind the knee.

The **posterior tibial artery** is the larger of the two branches of the popliteal artery. It runs down the back of the leg, deep to the gastrocnemius and soleus, alongside the tibial nerve. It gives off the **fibular artery** to the lateral leg. At the ankle it passes behind the medial malleolus, where its pulse can be felt, and enters the sole of the foot. The **anterior tibial artery** passes forward into the front of the leg and continues onto the top of the foot as the **dorsalis pedis artery**.

Veins of the lower limb

The lower limb has deep veins and superficial veins. The deep veins run beside the arteries and share their names. The posterior and anterior tibial veins join to form the popliteal vein. The popliteal vein becomes the **femoral vein**, which runs up the thigh beside the femoral artery. In the femoral triangle it lies on the medial side of the artery. Under the inguinal ligament the femoral vein becomes the external iliac vein. The deep veins carry most of the blood out of the lower limb.

The **great saphenous vein** is the main superficial vein and the longest vein in the body. It starts on the medial side of the top of the foot, passes in front of the medial malleolus, and runs up the medial side of the leg and thigh in the fat under the skin. Just below the inguinal ligament it passes through an opening in the deep fascia and empties into the femoral vein. Surgeons often use it as a graft in coronary artery bypass surgery. The smaller **small saphenous vein** runs up the back of the calf and empties into the popliteal vein.

THE LOWER LIMB VESSELS ON YOUR LIST

Vessel	Deep or superficial	Course and landmarks
Femoral artery	Deep	From under the inguinal ligament through the femoral triangle and down the medial thigh to the adductor hiatus

Vessel	Deep or superficial	Course and landmarks
Femoral vein	Deep	Beside the femoral artery, on its medial side in the femoral triangle; becomes the external iliac vein at the inguinal ligament
Great saphenous vein	Superficial	From the top of the foot, in front of the medial malleolus, up the medial leg and thigh to the femoral vein
Posterior tibial artery	Deep	From the popliteal artery down the back of the leg, deep to the calf muscles, to behind the medial malleolus

MEMORY TIP

In the femoral triangle, from lateral to medial, think NAVEL: Nerve, Artery, Vein, Empty space, Lymphatics. In the triangle the vein is medial to the artery.

Major nerves of the abdomen

Of the nerves in this part, only the femoral nerve is on your Module 4 structure list, and your nerve plexus notes also name the obturator nerve. The others are here so you can recognize what you see on the posterior abdominal wall of the cadaver.

The **lumbar plexus** forms inside the psoas major from the anterior rami of spinal nerves L1 to L4. Its branches come out of the psoas and cross the posterior abdominal wall. This is where you see them on the cadaver. They supply the lower abdominal wall, the groin, and the front, lateral, and medial thigh. Most of the anterolateral abdominal wall, from the ribs to below the umbilicus, is supplied by the lower intercostal nerves (T7 to T11) and the subcostal nerve (T12). The lumbar plexus supplies only the lowest strip, above the pubis and in the groin. The anterior and posterior vagal trunks, from the vagus nerve (cranial nerve X), enter the abdomen on the esophagus through the esophageal hiatus and supply the stomach and other abdominal organs.

BRANCHES OF THE LUMBAR PLEXUS ON THE POSTERIOR ABDOMINAL WALL

Nerve	Spinal nerves	Where you see it	What it supplies
Iliohypogastric nerve	L1	Crosses the posterior abdominal wall toward the iliac crest	The lower abdominal wall muscles and the skin above the pubis
Ilioinguinal nerve	L1	Runs just below the iliohypogastric nerve, then through the inguinal canal	The lower abdominal wall muscles and the skin of the upper medial thigh and the external genitalia
Genitofemoral nerve	L1 to L2	Runs down the front surface of the psoas major	The skin of the external genitalia and a small patch of the upper anterior thigh
Lateral femoral cutaneous nerve	L2 to L3	Crosses the iliacus toward the anterior superior iliac spine	The skin of the lateral thigh
Femoral nerve	L2 to L4	Comes out of the lateral side of the psoas major and runs in the groove between the psoas major and iliacus	The anterior thigh muscles and the skin of the anterior thigh
Obturator nerve	L2 to L4	Comes out of the medial side of the psoas major and runs to the obturator foramen	The adductor muscles of the medial thigh and the skin of the medial thigh

Major nerves of the lower limb

The **femoral nerve** is the largest branch of the lumbar plexus. It enters the thigh under the inguinal ligament, on the lateral side of the femoral artery. It supplies the anterior thigh muscles, including the quadriceps femoris and the sartorius, and the skin of the anterior thigh. One of its branches continues down to the skin of the medial leg.

The **sciatic nerve** comes from the **sacral plexus** (L4 to S3) and is the largest nerve in the body. It leaves the pelvis through the greater sciatic foramen, passes deep to the gluteus maximus, and runs down the back of the thigh deep to the hamstrings, which it supplies. Usually just above the knee it splits into two nerves.

The **tibial nerve** continues straight down through the back of the knee and the posterior leg, deep to the gastrocnemius and soleus, beside the posterior tibial artery. It supplies the posterior leg muscles (gastrocnemius, soleus, plantaris, tibialis posterior, flexor digitorum longus, and flexor hallucis longus). It passes behind the medial malleolus into the sole of the foot.

The **common fibular nerve** follows the tendon of the biceps femoris to the lateral side of the knee and wraps around the neck of the fibula, just under the skin. There it divides. Its superficial branch supplies the lateral leg muscles (fibularis longus and brevis). Its deep branch supplies the anterior leg muscles (tibialis anterior, extensor hallucis longus, extensor digitorum longus, and fibularis tertius). Because it lies so close to the surface at the fibular neck, a blow or a tight cast there can injure it and weaken the muscles that lift the foot, a problem called foot drop.

THE LOWER LIMB NERVES ON YOUR LIST

Nerve	Plexus	Course	Muscles supplied
Femoral nerve	Lumbar (L2 to L4)	Under the inguinal ligament, lateral to the femoral artery, into the femoral triangle	Anterior thigh: quadriceps femoris and sartorius
Sciatic nerve	Sacral (L4 to S3)	Greater sciatic foramen, deep to the gluteus maximus, down the back of the thigh deep to the hamstrings	The hamstrings; then splits into the tibial and common fibular nerves
Tibial nerve	Sacral, from the sciatic nerve	Back of the knee, deep to the calf muscles, behind the medial malleolus	Posterior leg muscles and the sole of the foot
Common fibular nerve	Sacral, from the sciatic nerve	Along the biceps femoris tendon and around the neck of the fibula	Lateral and anterior leg muscles, through its superficial and deep branches

CHAPTER 6

Endocrine Anatomy

ENDOCRINE

COMPETENCIES 128 TO 132

Covers the endocrine glands, where each one sits in the body, and the gross anatomy of the pituitary, thyroid, parathyroid, adrenal, and pancreatic glands. Hormone action belongs to physiology and is not covered here.

COMPETENCIES

1. **128** Major endocrine glands and locations
2. **129** Pituitary gross anatomy and lobes
3. **130** Thyroid and parathyroid anatomy
4. **131** Adrenal gland zones
5. **132** Pancreas, gonads, and organs with endocrine tissue

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Endocrine anatomy

drsrennie-stack.github.io/new-build-bio4-solano/endocrine-concept-videos.html

COMPETENCY 128 LECTURE

Major endocrine glands and locations

You should be able to: Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

The endocrine system is a scattered set of glands and tissues that release **hormones**, chemical messengers that travel in the blood. Unlike most organ systems, most of its glands are not physically connected to each other. The glands sit in the head, neck, thorax, abdomen, pelvis, and scrotum, and the blood links them.

An **endocrine gland** is a ductless gland. Its cells release their product into the surrounding interstitial fluid, and from there it enters the blood. Endocrine glands have a rich supply of capillaries for this reason. An **exocrine gland** has a duct that carries its product onto a surface or into a cavity, the way the salivary glands empty into the mouth. Some organs, such as the pancreas, have both kinds of tissue.

THE BASIC VOCABULARY

Term	What it is
Endocrine system	The glands and tissues that release hormones into the blood to signal distant cells
Hormone	A chemical messenger released by an endocrine cell that travels in the blood to target cells
Endocrine gland	A ductless gland that releases its product into the surrounding fluid and the blood
Exocrine gland	A gland that releases its product through a duct onto a surface or into a cavity; not part of the endocrine system
Target cell	A cell that carries the receptor for a given hormone and so responds to it

Learn the major glands from the top of the body down. Three are in the brain or hang from it: the **hypothalamus**, the **pituitary gland**, and the **pineal gland**. Two are in the neck: the **thyroid gland** and the **parathyroid glands** behind it. The

thymus is in the thorax. The **adrenal glands** and the **pancreas** are in the abdomen, and the endocrine tissue of the pancreas is the **pancreatic islets**. The **gonads** are the **ovaries** in the female pelvis and the **testes** in the male scrotum.

WHERE EACH GLAND SITS

Gland	Body region	Location
Hypothalamus	Brain	The lowest part of the diencephalon, below the thalamus; forms the floor and lower walls of the third ventricle. It links the nervous and endocrine systems and controls the pituitary
Pituitary gland	Brain	A pea-sized gland in the sella turcica of the sphenoid bone, hanging from the hypothalamus by the infundibulum
Pineal gland	Brain	A small, cone-shaped gland in the epithalamus, projecting from the posterior roof of the third ventricle
Thyroid gland	Neck	A butterfly-shaped gland in the anterior neck below the larynx; its two lobes lie beside the trachea and its isthmus crosses in front of it. The largest purely endocrine gland
Parathyroid glands	Neck	Usually four tiny glands on the posterior surface of the thyroid lobes, two on each side
Thymus	Thorax	Mediastinum, behind the sternum and above the heart; large in childhood, shrinks after puberty
Adrenal glands	Abdomen	Also called the suprarenal glands; one on the superior pole of each kidney, on the posterior abdominal wall behind the peritoneum
Pancreatic islets	Abdomen	Small clusters of endocrine cells scattered through the pancreas, which lies behind the stomach
Ovaries	Pelvis	Pelvic cavity, one on each side of the uterus
Testes	Scrotum	In the scrotum, outside the abdominopelvic cavity

MEMORY TIP

Learn the glands from top to bottom by body region. Brain: hypothalamus, pituitary, pineal. Neck: thyroid and parathyroids. Chest: thymus. Abdomen: adrenals and pancreas. Pelvis: ovaries, or testes in the scrotum.

WHERE A GLAND SITS SHAPES ITS DISEASE

The **pituitary gland** sits in the bony **sella turcica**, so a growing pituitary tumor usually expands upward out of the sella. Directly above it is the optic chiasm, and pressure there can cause a loss of peripheral vision. That vision change can be the first sign that a gland the size of a pea has enlarged.

The **thyroid gland** lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny **parathyroid glands** on its back surface must be carefully spared.

Each **adrenal gland** rests on the superior pole of a kidney. Imaging ordered for the kidney also captures the adrenal, which is why an adrenal mass is sometimes found by accident.

COMPETENCY 129 LECTURE AND LAB

Pituitary gross anatomy and lobes

You should be able to: Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

The **pituitary gland**, also called the **hypophysis**, is about the size of a pea. It sits in the **sella turcica**, a saddle-shaped pocket in the body of the sphenoid bone in the middle of the base of the skull. The bone forms the floor and the front and back walls of this pocket and protects the gland. The optic chiasm lies just above and in front of it.

The pituitary hangs from the hypothalamus by a slender stalk called the **infundibulum**. The infundibulum is the physical link between the brain and the gland. Nerve fibers from the hypothalamus run down through it, and small blood vessels run down along it.

THE PITUITARY AND ITS CONNECTIONS

Structure	What it is
Hypophysis	Another name for the pituitary gland
Sella turcica	The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary
Infundibulum	The slender stalk that connects the pituitary to the hypothalamus above it
Hypothalamo-hypophyseal tract	The bundle of nerve fibers that runs from the hypothalamus down through the infundibulum to the posterior lobe
Hypophyseal portal system	The blood vessels that carry hypothalamic signals directly to the anterior lobe

The two lobes

The pituitary has two lobes that develop from two different tissues in the embryo, so it is really two glands joined together. The **anterior lobe**, or **adenohypophysis**, grows up from the roof of the embryonic mouth. It is made of glandular epithelial tissue. Its cells are arranged in cords and clusters with wide capillaries running between them. The anterior lobe is the larger of the two.

The **posterior lobe**, or **neurohypophysis**, grows down from the floor of the brain. It is made of nervous tissue and is an extension of the hypothalamus itself. It contains the axons of neurons whose cell bodies sit in the hypothalamus, along with supporting glial cells called **pituicytes**. The posterior lobe makes no hormones of its own. It stores and releases what the hypothalamic neurons make and send down their axons.

Because the tissues differ, each lobe connects to the hypothalamus in a different way. The posterior lobe connects by nerve fibers, the **hypothalamo-hypophyseal tract**, which runs down through the infundibulum. The anterior lobe has no nerve link to the hypothalamus. It connects through blood vessels, the **hypophyseal portal system**.

1. **Primary capillary plexus** A capillary bed at the base of the hypothalamus, where the infundibulum begins, picks up hypothalamic signals.
2. **Hypophyseal portal veins** Small veins carry the blood down along the infundibulum.
3. **Secondary capillary plexus** A second capillary bed in the anterior lobe delivers the signals to the glandular cells.

THE LOBES OF THE PITUITARY GLAND

Feature	Anterior lobe	Posterior lobe
Other name	Adenohypophysis	Neurohypophysis

Feature	Anterior lobe	Posterior lobe
Tissue type	Glandular epithelial tissue	Nervous tissue, an extension of the hypothalamus
Develops from	An upgrowth of the roof of the embryonic mouth	A downgrowth of the floor of the brain
Link to the hypothalamus	Blood vessels, the hypophyseal portal system	Nerve fibers, the hypothalamo-hypophyseal tract, through the infundibulum
On a slide	Densely packed gland cells, many stained pink or blue, in cords with capillaries between them	Pale, fibrous-looking tissue of axons and scattered pituicyte nuclei

MEMORY TIP

"Neuro" means nervous tissue and "adeno" means gland. The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels.

COMPETENCY 130 LECTURE AND LAB

Thyroid and parathyroid anatomy

You should be able to: Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

The thyroid gland, gross anatomy

The **thyroid gland** is a butterfly-shaped gland in the anterior neck, just below the thyroid cartilage of the larynx. It has a **right lobe** and a **left lobe**. Each lobe lies against the side of the larynx and trachea. The two lobes are joined across the midline by a narrow bridge of thyroid tissue called the **isthmus**, which crosses in front of the upper rings of the trachea, just below the cricoid cartilage. A thin connective tissue capsule covers the gland. The thyroid is the largest gland in the body that is purely endocrine, and it has a very rich blood supply.

The thyroid gland, on the slide

Under the microscope, the thyroid is made of thousands of tiny hollow spheres called **thyroid follicles**. Each follicle is a ring of cells around a center filled with **colloid**. Colloid is a thick, protein-rich fluid. It stains pink and fills the center of each follicle.

The wall of each follicle is a single layer of **follicular cells**. They form a simple cuboidal epithelium that sits around the colloid. Follicular cells are the main hormone-making cells of the thyroid. Between the follicles, or tucked into the outer edge of a follicle wall, are the **parafollicular cells**, also called **C cells**. They are larger and paler than follicular cells, and they do not touch the colloid. They occur alone or in small groups.

THE THYROID, FROM THE GROSS GLAND TO THE FOLLICLE

Structure	What it is	How to find it
Right and left lobes	The two lateral parts of the gland	On each side of the larynx and trachea, below the thyroid cartilage
Isthmus	The bridge of thyroid tissue that connects the two lobes	Crossing the front of the upper trachea in the midline
Thyroid follicles	Microscopic hollow spheres that make up the gland	Rings of cells packed together across the slide

Structure	What it is	How to find it
Colloid	The protein-rich fluid stored in the center of each follicle	Pink material filling each ring
Follicular cells	The simple cuboidal cells forming the wall of a follicle	The single layer of cells lining each ring of colloid
Parafollicular (C) cells	Endocrine cells between the follicles	Larger, pale cells between follicles or at the outer edge of a follicle wall, not touching the colloid

The parathyroid glands

The **parathyroid glands** are small, round glands, each about the size of a grain of rice. Most people have four, a superior and an inferior gland on the posterior surface of each thyroid lobe. Each one has its own thin capsule that separates it from the thyroid tissue. The number and exact position vary from person to person.

Parathyroid tissue has no follicles and no colloid. Its cells are packed in cords and clusters with capillaries between them. Two cell types are seen on a slide. **Chief cells** (also called principal cells) are the small, numerous cells that make up most of the gland. They are the hormone-making cells of the parathyroid. **Oxyphil cells** are larger and much fewer. Their cytoplasm is packed with mitochondria, so it stains dark pink. Oxyphil cells become more common with age. Their function is not fully understood.

THE PARATHYROID GLANDS AND THEIR CELLS

Structure	What it is	How to find it
Parathyroid glands	Usually four small, round glands on the posterior surface of the thyroid lobes	Two on the back of each lobe, each in its own capsule
Chief cells	The main hormone-making cells of the parathyroid glands	Small, numerous cells with round nuclei and pale cytoplasm, making up most of the tissue
Oxyphil cells	Larger, fewer cells whose function is not fully understood	Larger cells with dark pink, granular cytoplasm, alone or in small groups among the chief cells

MEMORY TIP

Follicles and colloid mean thyroid. Packed cords of cells with no colloid mean parathyroid. On a parathyroid slide, the small, plentiful cells are chief cells and the big pink ones are oxyphil cells.

COMPETENCY 131 LECTURE AND LAB

Adrenal gland zones

You should be able to: Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

An **adrenal gland**, also called a **suprarenal gland**, sits on the superior pole of each kidney. Both adrenal glands lie on the posterior abdominal wall behind the peritoneum, enclosed with the kidney in fat and the renal fascia. The right gland is shaped like a pyramid. The left gland is shaped like a crescent. Each gland is covered by a connective tissue capsule.

Each adrenal gland is two glands in one. The outer **adrenal cortex** makes up most of the gland. It is glandular epithelial tissue that develops from mesoderm. The inner **adrenal medulla** is a small core in the center. It is modified nervous tissue that develops from the neural crest, the same embryonic tissue that forms the sympathetic ganglia. On a slide the cortex looks like layered bands and the medulla looks like a different tissue in the middle.

THE TWO PARTS OF AN ADRENAL GLAND

Structure	What it is
Capsule	The connective tissue covering of the gland
Adrenal cortex	The outer region and the bulk of the gland; glandular epithelial tissue arranged in three zones
Adrenal medulla	The inner core; modified nervous tissue that belongs to the sympathetic nervous system

The three zones of the cortex

The cortex is built in three zones. Each zone is named for how its cells are arranged. Follow them in order, from the capsule inward to the medulla.

1. **Zona glomerulosa** The thin outer zone, just under the capsule. Its cells form small rounded clusters and arches. "Glomerulus" means a little ball.
2. **Zona fasciculata** The thick middle zone, the widest of the three. Its cells line up in long, straight columns running toward the center. "Fascicle" means a bundle. The cells look pale and foamy because they hold many lipid droplets.
3. **Zona reticularis** The thin inner zone, against the medulla. Its cells form a branching network. "Reticulum" means a network. These cells stain darker than the fasciculata.

The adrenal medulla and chromaffin cells

The medulla is made of **chromaffin cells**. These are large, round to many-sided cells arranged in clusters around blood vessels. Chromaffin cells are modified sympathetic neurons. They have no axons or dendrites, and they release their product straight into the blood. They are supplied by preganglionic sympathetic nerve fibers, so the medulla works like a sympathetic ganglion. Their name comes from the brown color they turn when stained with chromium salts.

THE LAYERS OF THE ADRENAL GLAND, OUTSIDE TO INSIDE

Layer	Part of	How the cells are arranged
Capsule	Covering	Dense connective tissue
Zona glomerulosa	Cortex, outer	Thin layer of rounded clusters and arches
Zona fasciculata	Cortex, middle	Thick layer of long, straight columns of pale cells
Zona reticularis	Cortex, inner	Thin layer of cells in a branching network
Adrenal medulla	Medulla	Clusters of large chromaffin cells around blood vessels

MEMORY TIP

GFR, from the outside in: Glomerulosa, Fasciculata, Reticularis. Then the medulla, full of chromaffin cells, in the center.

COMPETENCY 132 LECTURE

Pancreas, gonads, and organs with endocrine tissue

You should be able to: Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

The pancreas and its islets

The **pancreas** is a long, soft gland in the upper abdomen. It lies behind the stomach, against the posterior abdominal wall and mostly behind the peritoneum. Its **head** sits in the C-shaped curve of the duodenum. Its **body** crosses behind the stomach, and its **tail** reaches toward the spleen on the left.

The pancreas is a mixed gland, with both exocrine and endocrine tissue. Most of it is exocrine tissue. This tissue is made of **acini**, clusters of cells that make digestive enzymes and send them through ducts to the duodenum. Scattered among the acini are about 1 to 2 million small islands of endocrine cells called the **pancreatic islets**, or **islets of Langerhans**. The islets have no ducts. They release their hormones into the many capillaries that run through them. On a slide, an islet is a round, pale-staining patch of cells surrounded by darker-staining acini.

The gonads

The **gonads** are mixed organs too. They make gametes, and they also contain endocrine cells. The **ovaries** are the paired female gonads. They sit in the pelvic cavity, one on each side of the uterus, against the lateral pelvic wall. Their endocrine tissue is the cells of the developing **ovarian follicles** and the **corpus luteum** that forms after ovulation. The **testes** are the paired male gonads. They sit in the scrotum, outside the abdominopelvic cavity. Their endocrine tissue is the **interstitial cells** (Leydig cells), which lie in the spaces between the seminiferous tubules.

THE MIXED ORGANS

Structure	Location	Endocrine part
Pancreas	Upper abdomen behind the stomach; head in the curve of the duodenum, tail toward the spleen	Both exocrine and endocrine
Pancreatic islets	Scattered throughout the pancreas	The endocrine cell clusters (islets of Langerhans)
Acini	Most of the pancreas	None; the exocrine clusters that make digestive enzymes
Ovaries	Pelvic cavity, one on each side of the uterus	Cells of the ovarian follicles and the corpus luteum
Testes	Scrotum	Interstitial (Leydig) cells between the seminiferous tubules

Other organs with endocrine tissue

Many organs whose main job is something else still contain endocrine cells. These organs are not counted as endocrine glands. Their endocrine cells still release hormones into the blood. The **heart** has hormone-making cells in the muscle of the atrial walls. The **kidneys** have endocrine cells in the renal cortex. The **skin** takes part too: cells of the epidermis begin making a hormone when the skin is exposed to sunlight. The stomach and small intestine, the liver, adipose tissue, and the placenta during pregnancy also contain endocrine cells.

ORGANS THAT CONTAIN ENDOCRINE TISSUE

Organ	Where the endocrine cells are
Hypothalamus	Neurosecretory cells; a neuroendocrine organ that controls the pituitary gland

Organ	Where the endocrine cells are
Heart	Cardiac muscle cells in the walls of the atria
Kidneys	Endocrine cells in the renal cortex, including the juxtaglomerular cells in the walls of the afferent arterioles
Skin	Keratinocytes of the epidermis, which begin making a hormone when exposed to sunlight
Stomach and small intestine	Enteroendocrine cells scattered in the mucosa
Liver	Hepatocytes, the main liver cells
Adipose tissue	Adipocytes (fat cells)
Placenta	Placental cells; a temporary organ of pregnancy

BIO 004 · Module 4

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